

Foreign Institutional Flow Regimes in Indian Equities

A complete account: motivation, construction, mathematics, validation, and economic meaning

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Long-form research document. Unlike the journal draft (FII_paper_draft.md), nothing here is compressed: every design decision carries its thought process, its mathematics, its reality check, and a citation to the code file, log, or table in this repository where the claim can be verified.

Citation convention. Code: `src/fii/...` (original Colab scripts preserved byte-for-byte in `legacy/colab_modules/`). Printed evidence: `outputs/validation/<stage>.log`. Tables/figures: `outputs/tables/`, `outputs/figures/`. Narrative history: `docs/research_log/`. Every number in this document regenerates via `python pipeline.py --all` (~8 minutes).

Part I — The question and the raw material

1. The question, and why anyone should care

1.1 The starting observation

Every trading day, India's financial press reports a single number: net foreign institutional investor (FII) buying or selling, in crores of rupees. It moves markets, headlines, and policy conversations. And it is, we will argue, close to the wrong number — because it collapses a *composition* into a *magnitude*.

Consider two days on which FIIs sell Rs.300 crore of the same stock. On the first, a single foreign desk unwinds a large position, demanding immediacy from the market. On the second, forty distinct institutions each trim the name slightly as part of broader portfolio adjustments. The headline number is identical. The economics could not be more different: the first is a *liquidity event* — a price concession extracted by someone who

needs to trade now — and standard microstructure theory (running from Kraus and Stoll’s 1972 block-trade study through Grossman–Miller and Campbell–Grossman–Wang) predicts the price impact should be partly *temporary*, reverting once the pressure stops. The second aggregates many independent decisions; if those decisions share an informational cause, the impact should be *permanent*.

The distinction is fifty years old in theory. What has been missing is the ability to *observe* it: public data reports flow magnitude, never flow composition. Nobody watching the daily FII number can tell a one-seller day from a forty-seller day.

1.2 The data opportunity

This project exists because of a dataset that makes composition partially observable: NSDL’s transaction-level records of FII trades in Indian equities, April 2011 – March 2025, with **masked but (within a month) distinct entity identifiers**. For every FII trade we observe the date, the ISIN, buy/sell, quantity, price — and a masked code for the foreign institution, its sub-account, and its broker. The masking destroys *who* is trading (deliberately, and, as §2 shows, far more thoroughly than we first assumed), but it preserves, within each day, *how many* distinct institutions participated and how the day’s volume was distributed across them.

That is exactly the measurement the fifty-year-old theory needs: a daily, stock-level observation of whether flow was **concentrated** (few entities, skewed participation) or **dispersed** (many entities, flat participation).

1.3 The hypothesis as originally posed — and a promise about honesty

We began (see docs/research_log/FII_Module1_findings.md, §0) with three behavioural archetypes, each grounded in a literature:

Archetype	Economic story	Expected flow signature	Literature anchor
Robot	passive / index-rebalance flow	transient, small uniform slices, low persistence	index/passive flow studies
Shark	informed conviction accumulation	persistent, blocky, concentrated book	Kyle (1985)-style informed trading
Hostage	forced fire-sale under redemptions	persistent selling, fragmented, dispersed book	Coval–Stafford (2007) fire sales

The pre-registered economic predictions were: Shark episodes should *drift* (informed flow anticipates returns), Hostage episodes should *reverse* (forced selling overshoots and recovers), Robot episodes should be *transient noise*.

Both directional predictions turned out to be wrong, and the final result is their inversion: concentrated flow (the “Shark”) behaves as liquidity demand and *reverts*; dispersed selling (the “Hostage”) behaves as information and is *permanent*. We flag this at the outset for two reasons. First, honesty: a reader should know the destination before judging the route. Second, methodology: the fact that the validation machinery was strong enough to overturn its designers’ expectations — twice, as Part IV recounts — is the strongest evidence we can offer that the machinery was not built to confirm anything. The archetype names were kept as fossils of the original hypothesis; the paper’s claims rest on what the data showed, not on what the names suggest.

1.4 Why a *regime* model rather than a daily signal

A natural first instinct is to compute a daily concentration number per stock and correlate it with returns. We rejected that design for three reasons, each of which shaped everything downstream.

First, institutional flow is a campaign, not a coin flip. Accumulation programs and liquidations are worked over days to weeks. Empirically, our persistence feature has lag-1 autocorrelation ≈ 0.93 (a 20-day window updated daily), and the fitted regimes dwell 13–17 trading days (outputs/validation/hmm_train_oos.log). A daily signal treats each day as independent and throws away exactly the temporal structure — onset, duration, exhaustion — that gives the events economic identity. The natural statistical object for “a persistent latent condition emitting noisy daily observations” is a hidden Markov model.

Second, event anchors need episodes. Our sharpest economic test (Part IV) compares returns *after flow stops* — the reversal test. “After it stops” is only defined if there is an “it”: an episode with a start and an end. Regimes give every stock-day a state, and every state run gives an END anchor date.

Third, discipline against data snooping. The regime model was built, frozen, and calibrated **entirely on flow data** — no prices, no index levels, no VIX, anywhere in the features (§3.6). Price data enters only afterward, as an out-of-domain validation. Had we optimized a daily signal against returns from the start, every downstream “finding” would be in-sample fit dressed up as discovery.

1.5 Why this is worth doing — the value question, round one

We will re-answer “why is this valuable” at every stage, as promised. At the stage of pure conception, the value proposition is: **a measurement that did not exist**. The transitory/permanent decomposition of institutional price impact is textbook theory with no daily-frequency, composition-based identification anywhere in the emerging-market literature, because the required data (entity-resolved institutional flow) essentially never leaves regulators and depositories. If the measurement works, everyone downstream of the daily FII number — execution desks deciding whether to trade into or around foreign flow, risk managers deciding whether a decline in a holding is noise or information, regulators deciding whether FII outflows warrant concern — gains a lens they currently lack. If it fails, the failure itself documents the limits of masked depository data. Either outcome is knowledge; that asymmetry is what justified the build.

2. The data and its brutal realities

2.1 The raw feed

Source. One parquet per year, data/ISIN_MAPPING/2011.parquet ... 2025.parquet (~450 MB compressed). Grain: one row per reported FII transaction. Key fields: FII, SUB_ACC, BRKER (masked entity IDs), ISIN, TR_DATE, TR_TYPE, RATE, QUANTITY, VALUE_INR, RFDE_INSTR_TYPE.

Scope filter (applied in src/fii/features/module1_feature_store_v2.py; rationale in docs/research_log/FII_Module §1): keep $TR_TYPE \in \{1 \text{ (buy)}, 4 \text{ (sell)}\}$ **and** $RATE > 0$ **and** $RFDE_INSTR_TYPE = "REG_DL_INSTR_EQ"$. This is not a convenience filter; it is a correctness filter. Transaction types 7/15/16/17 are corporate-action legs — bonus credits, split adjustments — and are 100% zero-rate in the feed; treating them as trades would fabricate enormous phantom “flow” on every corporate-action date. The instrument filter drops non-equity paper. What survives: **≈ 28 million** genuine equity buy/sell records. Unattributable rows (null/empty ISIN) were 278 of 24M in the v1 build — 0.001%, dropped.

Reality check that failed later, recorded here: we initially assumed the yearly parquets were one dataset among several; the ISIN-provenance audit (src/fii/validation/audits/module5g_isin_provenance.py,

log: outputs/validation/isin_provenance.log) later established the full funnel — **5,960** distinct raw FII-side ISINs → **3,812** canonical identities after ISIN-lineage collapse → **946** canonical ISINs (**939** true companies) in the final model universe. The 946 are the liquid subset: median 9,882 trades per name versus 623 in the excluded tail. The scope this creates — the model speaks about liquid large/mid-caps, ~25% of FII-traded names — is stated here once and inherited by every result in this document. (We originally hedged that fire-sale dynamics “may be stronger” in the excluded tail; Module 14 later *measured* the attributable tail and revised this: dislocations there are larger, but the reversal does not exist — see §13.6.)

2.2 The entity audit: the test that nearly killed the project

Everything interesting in this project runs through the masked entity IDs. Axis 3 — concentration — requires that “entity F387862664940” means *the same institution* across the observations we aggregate. Before building anything, we audited that assumption (docs/research_log/FII_Module1_findings.md §3). The audit ran three iterations because our first assumption about the ID format was itself wrong — a pattern that recurs throughout this project and that we came to treat as the normal epistemic state: **assume your assumptions are wrong until a test says otherwise**.

Step 1 — format heterogeneity. Character-shape profiling (map letters→A, digits→9) revealed *six* coexisting ID families, not the single F + 10 digits + YYYYMM pattern an initial sample suggested (that pattern covers only ~15% of rows):

Shape	Chars	Rows	Trailing YYYYMM matches report month?
A9999999999999	13	8.38M	no
A9999999999999	14	5.44M	no
A9999999999999999	17	4.29M	yes (~0.91)
A99999999999999999	19	2.16M	~0.77
literal "(null)"	6	1.73M	missing
true null	—	1.96M	missing

The scheme *changed over time*: an early era (~2011–2015) of 17/19-character IDs of the form PREFIX + core + YYYYMM (strip the last six characters to get identity), and a late era (~2021–2025) of plain 13/14-character IDs. Missingness is ~13% overall but heavily year-skewed: 0% in 2011 rising to **30% (2024) and 39% (2025)** — a coverage confound we carry as a stated limitation for all late-sample results.

Step 2 — the retention test and why it is insufficient. Define month-to-month retention as the share of month-*m* entities that reappear in month *m+1*. Measured with correct per-format identity, FII retention is 0.24–0.32. But retention *conflates two hypotheses*: “the institution didn’t trade next month” and “the institution’s mask was re-minted.” A low retention number cannot distinguish them.

Step 3 — the decisive test: months-per-entity, with a control group. The separation comes from a distributional statistic: for each distinct ID, count the number of distinct months it ever appears in, over the era’s full span. And crucially, run it on a population whose *true* persistence is known: **brokers**. Only ~150–320 brokers execute FII trades in any month — a near-fixed real-world set. If IDs were stable, broker IDs should appear in essentially *all* months of an 84–98-month span. The result:

Era (span)	Level	distinct IDs	median months	max months	≥12 months
Early (84 mo)	FII	95,987	2	10	0.0%

Era (span)	Level	distinct IDs	median months	max months	≥ 12 months
Early	BRKER (control)	12,700	2	10	0.0%
Late (98 mo)	FII	250,949	1	2	0.0%
Late	BRKER (control)	9,221	2	2	0.0%

A ~300-broker real set fragments into 12,700 (early) / 9,221 (late) distinct IDs, and **no ID of any kind, in any era, ever appears in twelve or more months**. The conclusion is unambiguous: *masked IDs are re-minted on roughly a monthly cycle; there is no stable cross-month entity identity in this data, at any level*.

What died, and what survived. Cross-month entity tracking died — we can never watch a specific fund liquidate over a quarter, which was the original Coval–Stafford-style conception of the Hostage. What survived: within a month, ID counts are sane (~150 brokers, ~3,000 FIIs per month), so IDs are internally consistent *within* the month, and therefore **within a day**. Axis 3 was rebuilt as a *single-day* concentration snapshot (§3.3), the Hostage was re-conceived as “a stock-day whose sell-side book is dispersed,” and the resulting loss of statistical power for that state was accepted and documented in advance (“the fragile state”). Two design consequences followed: entity level = FII (the umbrella best captures manager-level dispersal; SUB_ACC as robustness), and a per-stock-day **ID-coverage floor** (§3.3) so that days where too much value is unattributable are excluded rather than mismeasured.

Value at this stage, round two. The audit is a contribution independent of everything downstream: anyone who uses masked NSDL identifiers naively — and the temptation is strong, because the IDs *look* persistent — will manufacture spurious entity dynamics from mask re-minting. The months-per-entity-with-a-control-group design is the test we would hand any researcher facing a masked-panel dataset.

2.3 The base panel

From scoped trades, aggregation to the **(canonical ISIN, day)** grain (`src/fii/features/module1_feature_store_v2.py`):

$$BUY_{it} = \sum \text{VALUE_INR} [\text{TR_TYPE} = 1], \quad SELL_{it} = \sum \text{VALUE_INR} [\text{TR_TYPE} = 4],$$

plus buy/sell counts, total quantity, and the derived quantities

$$NET_{it} = BUY_{it} - SELL_{it}, \quad GROSS_{it} = BUY_{it} + SELL_{it}, \quad N_{it} = n_{it}^{buy} + n_{it}^{sell}, \quad \bar{ts}_{it} = GROSS_{it} / N_{it}.$$

Every feature in §3 is a function of these plus the entity-level values. The final store — `data/ISIN_MAPPING/stockday_features` — contains **2,423,212 stock-days across 3,812 canonical ISINs**, 2011-01-03 to 2025-03-28 (verified byte-identical on local regeneration: `outputs/diagnostics/fingerprints_after.json`).

Two structural facts about this panel drove the normalization strategy in §3.5: FII participation *breadth* grew ~254% over the sample, and there is a visible structural break around 2021 (a reporting-regime change). May–June 2021 is masked outright — removed before any trailing window is computed — so that no feature window spans the break; the same masked period later serves as the embargo buffer around the model’s train/test boundary (§6), which is not a coincidence but a design alignment.

3. Feature engineering: ten numbers per stock-day, and why exactly these

Full column-level documentation: docs/FII_stockday_data_dictionary.md. Construction code: src/fii/features/module1_feature_store_v2.py (stage feature_store; log outputs/logs/*_feature_store.log).

3.1 The unit of analysis, defended

The row is a **stock-day**. This was a contested choice worth defending explicitly (the full argument is §5 of the data dictionary):

1. **The label is a time-varying property of a stock.** “Reliance on 2020-03-23 looks like a fire sale” is a statement about a stock on a day. A static per-stock label or a per-entity label cannot express it.
2. **The phenomena are day-granular.** Block deals and redemption dumps are day-level events; the trades are day-dated; weekly aggregation would blur exactly the climax days the features exist to catch.
3. **It is the only grain that unifies stock-centric and entity-centric archetypes.** Robot and Shark are stories about flow into a name. The Hostage is a story about an *entity* (a fund dumping its whole book) — but given the audit’s verdict, entity trajectories are unobservable. The participation-weighted projection of entity book-concentration onto the stock-day (§3.3) is what lets all three archetypes inhabit one feature space. This projection is lossy, and the loss lands on the Hostage — the accepted, documented cost.
4. **Daily is not naively noisy**, because the panel is a hybrid: the output grain is daily, but two of the three model axes embed 20-day memory. A single wild day cannot dominate a stock-day’s coordinates.

One honest cost, flagged before modelling and revisited in §5: pooling ~929k complete-case stock-days into shared HMM emissions is *observation-weighted* — hyper-active large-caps contribute more rows, so fitted emissions tilt toward large-cap behaviour. Within-day ranking mitigates this; it does not remove it.

3.2 The three axes — the theory of the feature set

No single number separates three archetypes; separation requires position on three axes simultaneously:

- **Axis 1, persistence** — separates Robot (transient) from both persistent types. If flow direction is a campaign, its 20-day sign-consistency is high.
- **Axis 2, blockiness** — separates conviction (large prints) from fragmented execution (small uniform slices). Measured as *surprise against the stock’s own norm*, not in absolute rupees, to avoid price-level and liquidity confounds.
- **Axis 3, entity concentration** — the hard one and the point of the whole dataset: are today’s sellers (buyers) of this stock institutions acting *concentratedly* (their day’s book is mostly this name) or *dispersedly* (this name is a sliver of a broad book)?

3.3 The ten features, exactly as constructed

All formulas below are lifted from the construction code; the mapping table (raw → probit name) is at src/fii/features/module1_feature_store_v2.py, lines ~175–184.

Axis 1 — F_persist (persistence):

$$\text{pers_signed}_{it} = \left[\frac{1}{20} \sum_{k=1}^{20} \text{sign}(NET_{i,t-k}) \right], \quad \text{intensity}_{it} = \left[\frac{1}{20} \sum_{k=1}^{20} \frac{|NET_{i,t-k}|}{GROSS_{i,t-k}} \right],$$
$$\text{persistence_raw}_{it} = \text{pers_signed}_{it} \times \text{intensity}_{it} \in [-1, 1].$$

The sign-mean captures directional consistency; the intensity scaler damps stock-days where flow was two-sided (a +1 sign on a day that was 51% buys is not a campaign). Note the `shift(1)`: the window ends *yesterday* — today is never in its own baseline.

Axis 2 — F_block (blockiness surprise):

$$\text{blockiness_raw}_{it} = \frac{\overline{ts}_{it}}{\frac{1}{20} \sum_{k=1}^{20} \overline{ts}_{i,t-k}}.$$

Self-normalisation (today’s mean trade size over the stock’s own trailing mean) makes the feature price-level-neutral and stock-size-neutral: a Rs.2 cr print is blocky for a midcap and routine for Reliance.

Axis 3 — F_entity / F_entity_buy (entity book concentration, the dataset’s raison d’être). For entity e on day d , let $v(e, d, s)$ be e ’s sell value in stock s . The entity’s **book Herfindahl** that day:

$$HHI(e, d) = \sum_s \left(\frac{v(e, d, s)}{\sum_{s'} v(e, d, s')} \right)^2 \in (0, 1],$$

equal to 1 if the entity sold exactly one name and $\rightarrow 0$ as its selling spreads over many. The stock-day feature is the **participation-weighted average** over the entities selling s that day:

$$\text{entity_hhi_raw}(s, d) = \frac{\sum_e v(e, d, s) \cdot HHI(e, d)}{\sum_e v(e, d, s)}.$$

High $\rightarrow$ the sellers hitting this stock today are *focused* on it (concentrated selling); low $\rightarrow$ this stock is incidental to broad selling programs (dispersed). The buy-side analog (entity_hhi_buy_raw) is constructed identically on buy values. **Coverage gate:** the feature is null whenever less than 50% of the day’s sell value carries a valid entity ID — missing attribution must produce a missing feature, never a mismeasured one. A critical late audit finding: coverage is *value-weighted*, and even in 2024–25 (30–40% of rows missing IDs) attributable *value* clears the floor on essentially every stock-day, so Axis 3 survives the late sample.

Supporting features (context the three axes don’t span):

- **F_breadth:** n distinct entities trading the stock-day — crowdedness (Robot flow is broad). Later found to be the most characteristic-like feature (55% between-stock variance; §15).
- **F_sizedisp:** $\text{trade_size_std}/\overline{ts}$ — the coefficient of variation of print sizes; uniform VWAP-style slicing (Robot) vs lumpy discretionary prints (Shark).
- **F_activity:** $GROSS_{it}/GROSS_{i,t-1}^{20d}$ — activity surprise, a regime-onset marker. Disclosed redundancy: correlation 0.88 with F_block (both are “today vs own baseline” ratios).
- **F_streak:** signed length of the current run of same-sign NET days, as of *yesterday* (shifted). Correlation 0.56 with F_persist — related but distinct (a 3-day streak and a 15-day streak differ where the sign-mean may not).
- **F_imbal:** $NET_{it}/GROSS_{it}$ — today’s directional imbalance, the “standby axis.” Known artefact: probit ties at ± 1 (fully one-sided days) compress its tails.
- **F_flowbeta:** the trailing co-movement of the stock’s daily imbalance with the *aggregate* FII imbalance $\text{mkt_imb}_d = \sum_i NET_{id} / \sum_i GROSS_{id}$ — does this name ride the broad FII tide (Robot) or march to its own drum?

3.4 What was deliberately left out — and why that is load-bearing

No price. No index. No VIX. No macro. Anywhere. This is the single most important negative design decision in the project. The regime model’s later economic validation (Part IV) tests whether flow-defined states *predict price behaviour*. That test is only meaningful if price information had no path into the states. The features are flow-only, so the validation is out-of-domain by construction, not by assertion.

No flow-surprise (INNOV). A residual from a flow-expectation model (AAK-style) was designed early and *explicitly reserved as a validation object* (docs/research_log/FII_Module1_findings.md §2.1): if the same construct both fed the model and validated it, the validation would mark its own homework. This decision pays off in §14, where INNOV serves as the independent control that closes the last alternative explanation.

3.5 The normalization: within-day rank $\rightarrow$ probit, and the mathematics of why

Every raw feature is transformed, within each trading day d , over the eligible cross-section of n_d stock-days:

$$p_{it} = \frac{\text{rank}_d(x_{it})}{n_d + 1} \in (0, 1), \quad F_{it} = \Phi^{-1}(p_{it}),$$

where Φ^{-1} is the standard normal quantile (probit). Three distinct problems are solved by this one transform:

1. **Non-stationarity.** FII breadth grew 254% and the 2021 break shifted levels of nearly every raw feature. Cross-sectional ranking makes each day self-referential: F_{it} answers “*how unusual is this stock’s flow relative to today’s market*”, a question whose meaning is constant across fourteen years. Formally, if $x_{it} = g_t(\tilde{x}_{it})$ for any day-specific monotone distortion g_t (level shifts, scale drifts, reporting-regime changes), the rank — and hence F — is invariant to g_t . We buy stationarity by construction and pay by forfeiting market-wide level information (each day’s census shares are pinned by the quantile cut — an accepted cost, disclosed when interpreting census stability in §7).
2. **Emission honesty for the HMM.** Part II fits Gaussian emissions. The probability integral transform guarantees p_{it} is uniform on (0,1) under the within-day empirical distribution, hence $F_{it} = \Phi^{-1}(p_{it})$ has *exactly* standard normal marginals — no heavy tails, no skew, no outlier leverage. Fitting a Gaussian HMM to raw right-skewed HHIs or value ratios would hand the likelihood to a few extreme days; fitting it to probits is honest by construction. The empirical check (docs/FII_stockday_data_dictionary.md §4.3): post-probit means ≈ 0 (−0.042, 0.023, −0.001), standard deviations ≈ 0.97 , quartiles $\approx \pm 0.67$ — the theoretical $N(0,1)$ values.
3. **Outlier robustness.** Ranks are bounded; a data error in one raw value moves one rank slot, not a moment.

Eligibility and null semantics. Ranking runs over stock-days with $N \geq 5$ trades (the liquidity floor — a “concentration” measured off two prints is noise). A feature is null when the day is illiquid, in a stock’s ~15-day warm-up, or (Axis 3) below the coverage floor. Nulls are *honest missing values* that the model masks — a hygiene bug where warm-up NaNs masqueraded as present values (poisoning correlation and description statistics) was caught and fixed in the v1 build, recorded in the dictionary §4.3.

3.6 Feature validation: what we checked before any model saw the data

The feature store was validated *statistically* before modelling (economics comes later and is earned separately):

- **Marginals** $\approx N(0,1)$ as above (probit correctness, no tie artefacts).
- **Independence.** The three model axes are near-orthogonal — pairwise correlations $\rho(\text{persist}, \text{block}) = 0.015$, $\rho(\text{persist}, \text{entity}) = -0.142$, $\rho(\text{block}, \text{entity}) = 0.195$ — and the two non-zero terms point the economically *correct* way: concentrated players also print bigger blocks (+0.19); dispersed sellers skew sell-persistent (−0.14). Axes that were highly correlated would make the three-axis story an illusion of one axis.
- **Known redundancies disclosed, not hidden:** activity $\times$ block $r = 0.88$; streak $\times$ persist $r = 0.56$; imbalance tie-compression. These mattered later — the GBT challenger (§15) had access to all ten features precisely so that the “did the HMM leave signal behind” question included the redundant and standby features the HMM never used.

- **Sample accounting.** 2,423,212 stock-days total; ~1.15M pass the liquidity floor; **929,195 complete cases (80.7% of eligible)** on the three v1 axes; the ~19% gap decomposes into warm-up and coverage, both measured. The final complete-case store for the 10-feature v2 build feeds the model universe of §2.1 (946 canonical ISINs after a minimum-sequence filter of 60 stock-days per name, verified count-integral in outputs/validation/universe_audit.log... see Part II).

Value at this stage, round three. Even with no model attached, the feature store is infrastructure: a leakage-disciplined, stationarised, entity-audit-gated panel of FII flow microstructure across fourteen years. Every design element — the shift(1) convention, the mask-then-window rule, the value-weighted coverage gate, the probit marginals — is reusable by any successor project on this data, and each exists because a specific failure mode (lookahead, break-spanning windows, attribution decay, emission misspecification) was identified *before* it could contaminate results.

— *End of Part I. Part II (§§4–7) covers the HMM mathematics, the negative result that forced the hybrid architecture, the threshold calibration that falsified itself, and the descriptive statistics that quantify replication.*

Part II — The model

4. Why regimes, why an HMM, and the mathematics of hidden Markov models

4.1 The modelling bet, stated precisely

§1.4 argued informally that institutional flow is a campaign. The formal version of that bet: a stock’s daily flow-feature vector $\mathbf{x}_t \in \mathbb{R}^4$ is generated by a **latent state** S_t that (i) takes few values, (ii) persists over days, and (iii) determines the *distribution* of the observables, not their exact values. If that structure is real, a hidden Markov model recovers it; if it is not, the HMM’s own diagnostics — degenerate states, one-day dwell times, non-replicating signatures — expose the failure. Part of why we chose an HMM over softer clustering is exactly that it fails loudly.

4.2 The generative model

For each stock, an unobserved chain $S_1, \dots, S_T \in \{1, \dots, K\}$:

$$P(S_1 = i) = \pi_i, \quad P(S_t = j \mid S_{t-1} = i) = A_{ij},$$

with first-order Markov dependence (the future depends on the past only through the present state), and conditionally independent emissions

$$\mathbf{x}_t \mid S_t = j \sim \mathcal{N}(\mu_j, \Sigma_j), \quad \Sigma_j \text{ diagonal.}$$

Three model-class decisions and their reasons:

Gaussian emissions are honest here *only because of* the probit normalization (§3.5): each feature has exactly standard-normal marginals by construction, so a Gaussian emission family is not an approximation imposed on skewed data but a shape the data actually has.

Diagonal covariance ($\Sigma_j = \text{diag}(\sigma_{j1}^2, \dots, \sigma_{j4}^2)$) was chosen for rare-state estimability. A full covariance per state costs $\binom{4}{2} = 6$ extra parameters per state; a state occupying ~7% of the panel must estimate its covariance from proportionally few effective observations, and off-diagonal terms estimated from thin samples destabilize

EM. The cost is that within-state feature correlations are ignored — acceptable because the features were built near-orthogonal (§3.6: pairwise $|\rho| \leq 0.195$).

The feature subset. Only four of the ten features feed the HMM: `F_persist`, `F_block`, `F_entity_s`, `F_entity_buy_s` — one direct signature per hypothesized archetype (see docs/research_log/FII_Module2_hmm_log.md §0). `F_activity` was excluded for its 0.88 correlation with `F_block` (near-duplicate features double-count evidence in a likelihood), `F_streak` for its 0.56 correlation with `F_persist`; the rest were held in reserve. This exclusion decision is what later gives the LightGBM challenger (§15) its purpose: the GBT gets all ten, so “did the HMM’s four-feature diet leave predictive signal on the table?” becomes a testable question rather than a regret.

4.3 Likelihood, forward–backward, and Baum–Welch

The likelihood of an observed sequence marginalizes over all K^T state paths — naively intractable, and the reason the forward recursion exists. Define the **forward variable** $\alpha_t(j) = P(\mathbf{x}_{1:t}, S_t = j)$ and the emission density $b_j(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu_j, \Sigma_j)$. Then

$$\alpha_1(j) = \pi_j b_j(\mathbf{x}_1), \quad \alpha_{t+1}(j) = b_j(\mathbf{x}_{t+1}) \sum_{i=1}^K \alpha_t(i) A_{ij}, \quad P(\mathbf{x}_{1:T}) = \sum_j \alpha_T(j),$$

computed in $O(K^2T)$. The **backward variable** $\beta_t(i) = P(\mathbf{x}_{t+1:T} \mid S_t = i)$ satisfies the mirror recursion, and together they give the two posteriors EM needs:

$$\gamma_t(i) = P(S_t = i \mid \mathbf{x}_{1:T}) = \frac{\alpha_t(i) \beta_t(i)}{\sum_k \alpha_t(k) \beta_t(k)}, \quad \xi_t(i, j) = \frac{\alpha_t(i) A_{ij} b_j(\mathbf{x}_{t+1}) \beta_{t+1}(j)}{\sum_{k,l} \alpha_t(k) A_{kl} b_l(\mathbf{x}_{t+1}) \beta_{t+1}(l)}.$$

Baum–Welch is EM. With latent states, the complete-data log-likelihood is linear in the indicators $\mathbb{1}[S_t = i]$ and $\mathbb{1}[S_{t-1} = i, S_t = j]$; the E-step replaces them with their posteriors γ and ξ ; the M-step maximizes the resulting expected log-likelihood in closed form:

$$\hat{A}_{ij} = \frac{\sum_t \xi_t(i, j)}{\sum_t \gamma_t(i)}, \quad \hat{\mu}_j = \frac{\sum_t \gamma_t(j) \mathbf{x}_t}{\sum_t \gamma_t(j)}, \quad \hat{\sigma}_{jd}^2 = \frac{\sum_t \gamma_t(j) (x_{td} - \hat{\mu}_{jd})^2}{\sum_t \gamma_t(j)}.$$

Each iteration provably does not decrease the likelihood; convergence is to a **local** optimum. Two practical consequences we engineered around (`src/fii/models/hmm_stages/module3a_model_split_oos.py`): multiple random initializations (`N_INITS` restarts, keep the best log-likelihood; in the Module-2 fit two of five inits converged to the same optimum at $-1,424,681$ — reassuring), and a fixed seed per init so the entire fit is deterministic — which is why the local re-run of the pipeline reproduced the Colab census to within 3 rows in 804,958 (§7.4).

Decoding. Reported states are the Viterbi path — the single most probable state *sequence*, via dynamic programming on $\delta_{t+1}(j) = b_j(\mathbf{x}_{t+1}) \max_i \delta_t(i) A_{ij}$ with backtracking. Viterbi (rather than pointwise argmax of γ_t) respects the transition structure: it will not produce one-day state flickers that the transition matrix says are improbable — the correct choice when the downstream object is the *episode*.

Label switching. State indices are arbitrary (the likelihood is invariant to permuting states), so states are labeled **post hoc from their fitted signatures** — explicitly, the state whose mean `F_persist` is most negative is `SELL_REGIME`, most positive is `BUY_REGIME` (`module3a_model_split_oos.py` line ~101). Module 2’s

labeling code went further and *refused to bless* labels whose signatures contradicted the archetype definitions — a warning system that, as §5 recounts, fired immediately and correctly.

4.4 Fitting hygiene: the fit-cap and its own bug-fix

Pooling every stock-day into EM would let hyper-active large-caps dominate the emissions (§3.1). The mitigation: **fit balanced, decode everything** — each stock contributes at most 400 complete rows to the *fitting* sample (278,462 rows from 776,068 in the Module-2 fit), then the fitted model decodes all full sequences. The v1 implementation of the cap took each stock’s *most recent* 400 rows, quietly biasing emissions toward late-sample years; Module 3A replaced it with a **random contiguous block** of ≤ 400 days per stock (docs/research_log/FII_Module3_validation_log.md §1). Contiguity matters: EM learns the transition matrix from within-sequence adjacencies, so random *rows* would destroy exactly the temporal structure the model exists to estimate.

5. Why the HMM alone could not do it: the negative result that fixed the architecture

This section answers “why choose four regimes” — and the honest answer is that we did not choose them; we *discovered* the regime space’s true dimensionality by watching a sequence of HMMs fail in an informative pattern. The full record is docs/research_log/FII_Module2_hmm_log.md; the exploratory scripts are preserved in legacy/colab_modules/ (module2_v1 ... module2_v4).

5.1 The pre-registered criterion

Before the first fit, the success condition was written down: *the Hostage exists as a regime iff some fitted state shows clearly negative mean F_{persist} AND clearly negative mean F_{entity} — persistent, dispersed selling*. Pre-registering this is what prevented rationalizing whatever clusters EM happened to produce.

5.2 First fit: the model found direction, not archetypes

The 3-state fit (legacy/colab_modules/module2_v1_hmm_first_fit.py) produced states with mean F_{persist} of +1.17, +0.01, −1.14 — a **buy / neutral / sell triad** — and the labeling code’s contradiction warnings fired on both directional states. The reason is visible in the magnitudes: persistence means of ± 1.15 dwarf entity means of ± 0.2 , so the likelihood surface is dominated by one axis. Two economically real facts were extracted from the failure: the persistent-sell state’s *sellers were concentrated on average* (+0.17) — meaning it mixed dispersed and concentrated selling rather than separating them — and dispersed selling co-occurred with persistent *buying* (−0.22 in the buy state): broad-book entities are the liquidity **providers** during accumulation. Dispersion alone is not distress; the Hostage requires dispersion *and* sell pressure jointly.

5.3 Dissection and the k-sweep: the mixture exists, EM will not carve it

Was the material even there? Dissecting the 220,481 persistent-sell stock-days: 25.8% had dispersed books ($F_{\text{entity}} < -0.5$), 36.8% concentrated ($> +0.5$) — a genuine mixture the model was not separating. A BIC sweep over $k = 2 \dots 6$ (legacy/colab_modules/module2_v2_dissection_bic_sweep.py) then showed why more states would not help. On $n \approx 2.7\text{M}$ observations, BIC ($-2 \log L + p \log n$) declines monotonically in k — the big- n regime where the $\log n$ penalty is negligible against any real likelihood gain — with a true elbow at $k = 5$. But the $k = 5$ and $k = 6$ solutions were a **persistence ladder** (state means −1.52,

−0.66, ≈0, ≈0, +0.66, +1.54 on one axis, plus one-day transient noise states): finer gradations of direction, never a persist<0 and entity<0 state. The pre-registered criterion failed at every k .

5.4 The mechanism, diagnosed before re-engineering

Why is the entity axis invisible to EM? The hypothesis: **an HMM state must be temporally coherent** — EM allocates states along features whose values persist, because persistent features are what make the Markov chain’s self-transition probabilities informative. The test — median per-stock lag-1 autocorrelation of each input (`legacy/colab_modules/module2_v3_autocorr_smoothing_refit.py`):

feature	lag-1 autocorr
F_persist	0.926
F_block	0.290
F_entity (single-day snapshot)	0.334
F_entity_buy	0.295

We had fed the model one ultra-smooth feature and three jittery ones. The entity-audit’s single-day constraint (§2.2) had not merely weakened Axis 3 — it had made it *structurally invisible to this model class*.

The fix, and why it is audit-legal: F_entity_s, F_entity_buy_s = 5-day trailing means of the *daily HHI snapshots*, re-ranked within day and re-probited. This averages stock-level dispersion **measurements** across days; it requires no cross-day entity identity whatsoever, so it lives within the audit’s constraint. Lag-1 autocorrelation rose to 0.767/0.763, and the entity axis became HMM-visible.

5.5 The $k=4$ fit: the regime space is two-dimensional

The refit sweep on the smoothed features produced the single most informative fit of the project. At $k = 4$, EM kept the two directional poles and split the *neutral* middle by **participant structure**: a concentrated-both-sides state (entity means +0.56/+0.78, dwell 14 days) versus a dispersed-both-sides state (−0.45/−0.67, dwell 9.5 days). The regime space is **flow direction** × **participant concentration** — genuinely two-dimensional — and the four *occupied* cells were sell/concentrated, neutral/concentrated, neutral/dispersed, buy/mildly-dispersed. The two **empty corners were exactly the original hypotheses**: sell/dispersed (the Hostage) and buy/concentrated (the accumulating Shark). EM allocates states by probability mass; rare corners never earn one.

Hence the Module-2 headline finding, which the pre-registered criterion certified at every k , both raw and smoothed:

Persistent selling in Indian FII flow is, on average, concentrated-seller selling. The Coval–Stafford Hostage — persistent *and* dispersed selling — does not exist as a temporal regime at the stock-day level. Hostage days exist (~26% of persistent-sell days), but they are **episodic**, not regime-forming.

5.6 The hybrid architecture — and why “four archetypes” is the honest count

The pre-registered fork for this outcome: use the HMM for what it *can* see (temporal flow direction) and overlay deterministic rules on the entity coordinates for what is episodic (`legacy/colab_modules/module2_v4_final_hybrid_ov` production implementation `src/fii/models/hmm_stages/module3a + module3b`):

Label	Rule
HOSTAGE	SELL_REGIME and $F_{\text{entity}_s} < \text{threshold}(-)$
SHARK_DIST	SELL_REGIME and $F_{\text{entity}_s} > \text{threshold}(+)$
SHARK_ACC	BUY_REGIME and $F_{\text{entity}_\text{buy}_s} > \text{threshold}(+, \text{buy})$
ROBOT	NEUTRAL backbone state
UNTAGGED_DIRECTIONAL	directional regime, unremarkable entity structure

So the direct answer to “why four regimes”: **three** is the backbone count (the only temporal structure the data supports — direction), **four** is the archetype count because concentration adds one binary distinction on each directional pole that the data occupies (concentrated-sell, dispersed-sell, concentrated-buy; dispersed-buy never formed even an episodic cluster), and there is deliberately a **fifth non-label** — UNTAGGED, ~30–36% of stock-days — the honest “directional but unremarkable” bucket that prevents force-fitting every day into a story. Architectures that lack an “unremarkable” class manufacture archetypes out of noise.

Economically, the hybrid is arguably *trueer* than the original design: fire sales are **episodes within selling regimes**, not regimes themselves — a fund’s redemption dump lasts days, embedded in longer market-level selling conditions.

Post-review addendum — the ablation that settles the HMM’s role. A referee asked the right question: is the HMM central, or sophisticated machinery on a simpler idea? We tested it (`src/fii/validation/module13a_backbone_ablation`): a three-line rule backbone — TRAIN-era F_{persist} quantile cuts ($-0.510/+0.539$), census-matched to the HMM so both tag the same fraction of days — with the identical frozen overlays, reproduces the paper’s main table within band on every cell (rule SHARK_DIST $+71.6/+47.4$ vs HMM $+65.4/+48.5$; SHARK_ACC $-78.0/-52.3$ vs $-87.9/-47.6$; label agreement 92.7%, Cohen’s $\kappa = 0.891$). What the HMM buys is **episode smoothness, not identification**: rule-based runs are ~14% shorter (no transition-matrix persistence prior), producing ~48% more episode-ends with the same economics. The honest conclusion, adopted throughout this document: *the contribution is the participant-composition measure; the HMM is a taxonomy convenience* — retained for its cleaner episode segmentation, claimed for nothing more. The finding is model-class-independent.

5.7 Face validity: do the labels find real events?

Before any price data touched the project, the longest HOSTAGE episodes were mapped to companies and dates (`docs/research_log/FII_Module2_hmm_log.md` §6). The top of the list: JSW Steel, 53 days, starting seven days after Bernanke’s May 2013 taper speech; Vedanta, 42 days, during the 2015 commodity crash; HDIL and Reliance Communications — **both later bankrupt** — in their documented distress windows; Ambuja Cements in the demonetization aftermath. Two reads: the episodes land on canonical macro-stress windows *despite* within-day ranking removing the market-wide component (these are the names sold hardest relative to an already-stressed market — precisely Coval–Stafford’s prediction that fire sales concentrate in commonly-held names), and the label finds genuine distress names years before their failures. Face validity is not proof — it is a necessary condition that, had it failed, would have ended the project cheaply.

6. Freezing, thresholds, and the calibration that falsified itself

6.1 The temporal protocol

Everything downstream rests on one discipline (`src/fii/models/hmm_stages/module3a_model_split_oos.py`; frozen dates in `config/config.yaml`): **train = stock-days $\leq$ 2021-04-30; test = $\geq$ 2021-07-01; May–June 2021 masked** (the same window already excised from the features — the structural break doubles as the embargo buffer, so no trailing window and no fitted parameter ever touches the boundary). HMM parameters, state identities, and overlay thresholds are derived on TRAIN only, frozen, and applied unchanged to TEST. The test era is not “more data”: it is 295,773 stock-days across **830 stocks versus TRAIN’s 572** — a ~45% larger and substantially different universe, which makes replication a real test rather than an echo.

6.2 Why ± 0.5 had to die

Module 2’s overlay threshold (± 0.5 probit $\approx$ 69th percentile) was an eyeballed placeholder. A threshold defines the archetypes, hence every downstream event study and regression; leaving it arbitrary would put an unexamined free parameter under the whole edifice. Module 3B’s design (`src/fii/models/hmm_stages/module3b_threshold_calibration.py`) fit a BIC-selected Gaussian mixture to the distribution of F_{entity_s} within the TRAIN sell regime, take the posterior-0.5 boundary of the dispersed component as the HOSTAGE threshold — **with two falsification checks pre-registered**: a k-means cross-check, and a stability requirement that re-deriving the boundary on TEST move it by less than 0.25.

6.3 What happened: the calibration failed its own test

The TRAIN GMM selected $k = 2$ — but with weights **49%/51%** at means $-0.50/+0.84$: a coin-flip bisection of the sell regime, not the rare-tail-versus-majority structure a fire-sale component should have. Its boundary, **+0.155**, would have labeled mildly *concentrated* days as HOSTAGE, doubled the Hostage census to ~14%, and eliminated the honest middle (every sell-regime day would have become either HOSTAGE or SHARK_DIST). Every independent check disagreed with it *and agreed with each other*: the k-means boundary was **-0.441**; the TEST-era GMM found a proper rare tail (weight 0.24, mean -0.84) with boundary **-0.510**; the stability delta, 0.665, tripped the pre-registered UNSTABLE flag. The fallback engaged per protocol: **the quantile rule** — the TRAIN 25th percentile of F_{entity_s} within the sell regime:

$$\text{thr}_H = q_{0.25}(F^{\text{entity}_s} \mid \text{SELL, TRAIN}) = -0.513,$$

with the symmetric $q_{0.75}$ rules giving SHARK_DIST $> +0.877$ and SHARK_ACC $> +0.795$. The three-way convergence — k-means -0.441 , test-GMM -0.510 , train-quantile -0.513 , all within 0.07 of each other and of the original eyeballed -0.5 — is what elevates the threshold from “assumed” to “derived, with convergent evidence from methods with different assumptions.”

Per-threshold stability of the final rule (train- vs test-quantile): HOSTAGE -0.513 vs -0.331 ($\Delta 0.18$, and the drift direction carries the 2024–25 ID-coverage fingerprint, §2.2); SHARK_DIST $\Delta 0.09$; SHARK_ACC $\Delta 0.003$. The calibrated census restores the honest middle: ROBOT 42.9/43.6%, UNTAGGED 35.5/35.9%, SHARK_DIST 7.3/8.0%, SHARK_ACC 7.0/7.0%, HOSTAGE 7.3/5.6% (train/test; the Hostage test-dip is the coverage confound, not economics). Full numbers: `outputs/validation/threshold_calibration.log`, `outputs/tables/T2_archetype_census.csv`.

Did the calibrated threshold “improve accuracy”? The right claim is sharper: it prevented a *specific, quantified* mislabeling (a $+0.155$ cut mislabels mildly-concentrated days as dispersed fire-sales and doubles the HOSTAGE census with days that contradict its definition), and the key downstream statistic proved robust to

the choice — the episode-clustering ratio (§7.3) held at $\sim 2.6\times$ even under the rejected threshold. We therefore do not claim the threshold manufactured the result; we claim it made the labels mean what their definitions say.

6.4 A provenance lesson learned in this repository

One engineering honesty note. The original Colab session *observed* the UNSTABLE flag and applied the quantile fallback manually; the script as saved only printed the warning. When this pipeline was first re-run locally, the calibration silently regenerated the *bad* GMM thresholds — caught because the archetype census diverged from the frozen fingerprint (outputs/diagnostics/fingerprints_before.json). The falsification rule is now **binding in code** (module3b_threshold_calibration.py: $\Delta \geq 0.25 \rightarrow$ quantile rule for all three thresholds), and the local re-run reproduces the frozen thresholds to the third decimal. The episode is preserved because it is the project’s thesis in miniature: decisions that live in a person’s memory instead of in code are not reproducible, and only a fingerprint gate caught the difference.

7. Descriptive statistics and out-of-sample replication: quantifying “accuracy” for an unsupervised model

7.1 What “accuracy” can even mean here

A supervised model has ground truth; an unsupervised regime model does not — there is no oracle file of true fire-sale days. “Accuracy” must therefore be decomposed into falsifiable sub-claims, each with its own test:

1. **Structural stability** — do the fitted states mean the same thing on unseen data? (§7.2)
2. **Statistical reality of the labels** — do labeled days cluster in time like genuine episodes rather than i.i.d. noise? (§7.3)
3. **Reproducibility** — does an independent re-run produce the same labels? (§7.4)
4. **External validity** — do the labels align with events (face validity, §5.7) and with independent constructs (PIN, Part IV §16)?
5. **Economic content** — do the labels *predict price behaviour out of sample*? — the entirety of Part IV, and the only test that can make the labels matter.

The claim “the model is 93% accurate” is unavailable and would be meaningless; the claim “every measurable property of the model replicates on a disjoint era and a 45% different universe” is available, and is what this section documents.

7.2 Backbone replication (log: outputs/validation/hmm_train_oos.log)

The frozen TRAIN-fitted model, decoding both eras:

Check	TRAIN	TEST (frozen parameters)
SELL signature (persist / entity_s)	−1.04 / +0.18	−1.07 / +0.30
BUY signature (persist / entity_s)	+1.08 / −0.299	+1.13 / −0.304
Census (sell / neutral / buy)	29.1 / 42.9 / 28.0	28.5 / 43.6 / 27.9
Transition diagonal	0.95 / 0.95 / 0.95	0.95 / 0.95 / 0.95

Signatures, occupancy, and transition structure replicate almost digit-for-digit across disjoint years on a substantially different stock universe. One caveat we impose on ourselves: census stability is partly *by construc-*

tion (within-day ranking pins daily cross-sectional shares), so the informative rows are the signatures and transitions, which ranking does not pin.

7.3 Effect sizes and the permutation test (log: outputs/validation/model_descriptives.log)

With $n \approx 800,000$, p-values are decoration — everything is “significant.” The battery is therefore **effect-sizes-first**, using Cohen’s $d = (\bar{x}_1 - \bar{x}_2) / s_{pooled}$:

Contrast	F_persist	F_block	F_entity_s	F_entity_buy_s
HOSTAGE vs SHARK_DIST	+0.22 → +0.26	−0.19 → −0.38	(tautological)	−0.46 → −0.55
SHARK_ACC vs ROBOT	+3.19 → +3.20	~0	~0	+1.52 → +1.42
HOSTAGE vs ROBOT	−2.98 → −3.01	−0.19 → −0.34	−1.40 → −1.43	−0.74 → −0.71

(Arrows: TRAIN → TEST.) The flagged caveat comes first: the HOSTAGE-vs-SHARK_DIST gap on F_entity_s itself ($d \approx -5.7$, KS $D = 1.0$) is **tautological** — those labels are *defined* by a cut on that variable — so it is excluded from evidence. What counts: (i) every non-tautological effect size moves by ≤ 0.15 between disjoint eras — the label structure generalizes; (ii) within the sell regime, dispersed and concentrated sellers are near-identical on persistence and blockiness — the entity axis carries independent information, as designed; (iii) a small, correctly signed bonus: HOSTAGE prints are *smaller* than SHARK_DIST prints (block $d = -0.19/-0.38$) — fire-sale fragmentation versus block distribution, exactly the original framework’s sign, honestly reported as small.

The episode-clustering permutation test asks: do HOSTAGE days chain into episodes, or would i.i.d. tagging at the same rate produce the same runs? Null distribution: 200 within-stock shuffles of the label sequence — preserving each stock’s label *count* while destroying temporal adjacency — with mean run length as the statistic; $p = (1 + \#\{null \geq obs\}) / (1 + B)$:

Era	Observed mean run	Shuffled null	Ratio	p
TRAIN	3.64 d	1.47 d	2.48×	0.005
TEST	3.39 d	1.42 d	2.38×	0.005

Replicating out of sample, and robust to the threshold choice ($\approx 2.6\times$ even under the rejected +0.155 cut). Episode-length distributions are economically sensible: the active archetypes come in 3-day-median bursts with heavy tails (max HOSTAGE run: 53 days — the JSW Steel taper-tantrum event); ROBOT dwells for weeks (median 11 days, max 854).

7.4 Reproducibility as a measured quantity

The entire chain — features, HMM fit, calibration — was re-executed from raw data on different hardware, a different OS, and a different numpy/ hmmlearn generation than the original Colab runs (outputs/diagnostics/fingerprints_after.json): the feature store regenerated **byte-identical** (2,423,212 rows / 3,812 cisins); the calibrated archetype census matched **exactly on all four active archetypes** (HOSTAGE 53,613; SHARK_DIST 60,711; SHARK_ACC 56,275), with three stock-days in 804,958 flipping between the two control categories (ROBOT/UNTAGGED) — EM tie-breaking at machine

precision, 4×10^{-6} of the panel; and the downstream headline regression reproduced to ± 0.1 bp (§12). For an unsupervised pipeline with an EM fit inside it, this is the strongest available form of the “accuracy” claim: **the model is a deterministic function of the data and the frozen protocol, not of the session that happened to produce it.**

Value at this stage, round four. Even a reader who rejects every economic claim in Part IV inherits, from Parts I–II: an audited masked-ID dataset; a stationarised, leakage-disciplined feature store; the negative result that single-day concentration snapshots are HMM-invisible (with the autocorrelation diagnosis and the audit-legal smoothing fix — directly reusable by anyone modelling regime structure on jittery microstructure features); the demonstration that the regime space of institutional flow is direction $\times$ concentration with empty rare corners; and a calibration protocol whose falsification test fired and was survived. These are methods-contributions independent of whether concentrated FII selling reverts.

— *End of Part II. Part III (§§8–9) builds the price panel: the false assumption about NSE’s prev_close, the corporate-action mathematics and its gates, the symbol-migration guard, and the issuer-bounded ISIN closure that recovered 8% of the panel.*

Part III — Building prices you can trust

The regime model never touched price data; the economics of Part IV depend entirely on it. This part documents why “just download the prices” consumed more engineering and more failure-recovery than the model itself — and why that was the correct allocation of effort. The unabridged incident record is docs/research_log/FII_Module5_validation_log.md §§1–3e.

8. The price panel: repairs, corporate-action mathematics, and gates

8.1 What the price data must support, and the central hazard

The pre-registered economic tests need, for every stock in the model universe, a daily return series that is: **survivorship-free** (fire-sale stocks delist — dropping them deletes exactly the HOSTAGE worst cases), **corporate-action-true** (a 1:5 split prints -80% on the tape that no investor experienced), and **identity-continuous** (§9). The only available source is the NSE bhavcopy — raw daily closes, 2011–2025, including delisted names (data/VALIDATION_DATA/bhavcopy_parquets/).

The central hazard, stated numerically: on known split/bonus ex-days, the raw tape’s median absolute “return” is **0.508** — the signature of factor-2 events (the modal case: a price halving that is bookkeeping, not economics). A single unadjusted split inside a 20-day measurement window fabricates a crash larger than any effect we are testing for by two orders of magnitude (50,000 bp of artifact against ~ 50 bp of hypothesized signal). Most of Module 5’s effort was neutralizing this hazard *verifiably*.

8.2 Panel assembly and the audit-driven repairs

src/fii/data_prep/module5a_price_panel.py (stage price_panel) assembles 5.95M stock-days from 15 yearly parquets. Each repair was triggered by a specific finding of the raw-data audit (src/fii/validation/audits/module4c_data_audit.py):

- **R1 — the year-0020 bug.** 1,703 rows in prices_2020 carried literal year 0020 (a two-digit-year parse upstream). Poisonous precisely because date joins *silently drop* such rows — no error, just missing data. Repair: offset +2000 years.
- **R2 — the 2011 null-ISIN hole.** All early-2011 rows (169,866) had null ISINs. Repair: backfill each symbol’s ISIN from its next observed value; 1,341 unfillable rows dropped. This repair carried known

risks (symbol recycling, CA-timing) that were *not* waved away: it was validated only later and independently, when the canonical v3 panel showed 2011 coverage of 98.45% (§9.4) — had the backfill been wrong, that number would have cratered.

- **R3 — series dedupe.** One (isin, date) can appear under multiple NSE series; keep EQ > BE > BZ.
- **Macro timezone repair.** yfinance NIFTY/VIX dates landed one day early (20% on weekends — a tz artifact): +1 day, after which 3,491/3,492 dates matched the trading calendar. The S&P 500 enters lagged one day (US closes after India — a lookahead otherwise).

Two gates certified the assembly. **G1 (survivorship):** RCOM, HDIL, DHFL, JETAIRWAYS and peers present with correct delisting-era end dates — the panel is survivorship-free. **G2 (join coverage):** model stock-days match price rows at 90.7%, and — the part that matters for inference — **uniformly across archetypes (89.8–91.6%)**: no archetype-selective attrition, so measurement error is not correlated with the groups being compared. (Where the missing 9.3% went, and why a crosswalk was *not* the answer, is §9.2.)

8.3 Failure #1: the assumption that died on a gate

The original design assumed NSE’s `prev_close` column is restated on ex-dates — i.e. that `close/prev_close - 1` is already CA-adjusted. This assumption was **tested rather than trusted** (Gate 0, in the retired `legacy/colab_modules/module5b_forward_returns.py`): on 672 known split/bonus ex-days, median `|ret|` from `prev_close` = 0.505 $\approx$ median `|ret_cc|` = 0.506 from close-to-close. Had `prev_close` been restated, the two medians would diverge wildly on exactly those days; their identity proved **prev_close is raw**, voided every downstream number computed to that point, and forced the adjustment to be *built*, not assumed. The episode also produced a process correction that shaped the whole repository: the retired script had bundled the data gate and the model-facing analysis in one step, and the user’s correction — *one verifiable step per script* — became the stepwise, gated architecture the pipeline still has.

8.4 The corporate-action mathematics

For a face-value split $f_{old} \rightarrow f_{new}$ and a bonus of $a:b$ (a new shares per b held), the ex-date price divisors are

$$F_{split} = \frac{f_{old}}{f_{new}}, \quad F_{bonus} = \frac{a+b}{b}, \quad F_{same-day} = F_{split} \cdot F_{bonus},$$

(e.g. TITAN 2011: a 10→1 split plus 1:1 bonus $\rightarrow F = 10 \times 2 = 20$), and the adjusted return over an ex-date is

$$r_t^{adj} = (1 + r_t^{cc}) F - 1.$$

Worked example from the log: a raw -94.7% print on a $\times 20$ day becomes $0.053 \times 20 - 1 = +6.6\%$ — the stock actually *rose* that day.

Parsing and tape-verification (`src/fii/data_prep/module5b1_ca_factors.py`, stage `ca_factors`). Factors are parsed from the NSE corporate-action file’s free-text PURPOSE field — and every parsed factor is then **verified against the tape**: the observed ex-day ratio P_{t-1}/P_t must equal the factor within a $\times/\div 1.3$ band (the band absorbs the genuine same-day market move). This verification is what caught **Failure #2, the combined-event parser bug**: on rows like “*Bonus 1:1 And Face Value Split Rs.10 To Re.1*”, the v1 split regex grabbed the first two numbers in the string — the bonus ratio — silently dropping the split component on TITAN, ONGC, NALCO and peers. The tell was diagnostic, not lucky: the disagreement table clustered at `obs/factor` ≈ 5 and ≈ 10 , exactly the missing split magnitudes. The v2 parser anchors split numbers to text

after the split keyword and deduplicates to at most one split plus one bonus per symbol-day. Result: 570/570 splits parsed, **99.1% of 1,103 factors tape-confirmed** (splits 99.5%, bonuses 98.8%), and each of the 7 residual disagreements individually explained and *correct to fail* — e.g. DRREDDY/NTPC “bonuses” that were **debenture** issues with no equity price effect (applying them would fabricate –50%/–86% crashes), and postponed ex-dates. Policy locked: apply confirmed or unverifiable factors; null (never adjust) the ex-day return of confirmed-wrong events.

8.5 Failure #3: the silent clipboard corruption — and the process fix

The application step (`src/fii/data_prep/module5b2_apply_adjustment.py`, stage `apply_adjustment`) initially produced internally impossible output: median `|ret_adj|` = 2.04, zero ex-days small, and TITAN’s ex-day `ret_cc` printed +0.053 on the same row where step-1 had measured an 18.75× ratio. Numbers that contradict each other within one printout mean the *executed code* differs from the file. A read-only diagnostic (`src/fii/validation/audits/module5b2d_diagnose.py` — deliberately diagnostic-before-fix) found the smoking gun: the stored `ret_cc` equalled a fresh recompute **plus exactly 1.0 on all 5.94M rows**. The hand-pasted Colab cell had lost the “–1” from the return formula — clipboard corruption that, unlike incident #1 (a loud `SyntaxError`), ran silently. The file’s formula was correct all along.

The process fix outlived the incident: **scripts became the only executable artifact** — stored in Drive, run byte-exact via `exec(open(path))`, never pasted — which is the direct ancestor of this repository’s runner (`src/fii/runner.py`) executing preserved stage files. The same diagnostic run also vindicated the factor table independently (applying factors to the *fresh* series: median `|adj|` = 0.037) and surfaced two structural findings: splits **mint new ISINs** (breaking per-ISIN return chains at exactly the biggest corporate events — the motivation for §9), and **symbol migrations** (Failure #4: UNOMINDA’s split occurred under its old symbol MINDAIND, so the CA row pointed at a price chain not containing the event — even correct code would have fabricated a +2860% return). The migration class was closed with an **application guard**: any factor-day still showing `|ret_adj| > 50%` after adjustment is nulled and logged. The guard caught 45 days — on inspection, all renames (ZYDUSLIFE ex-Cadila, TATACONSUM ex-Tata Global, CYIENT ex-Infotech...).

8.6 The certification gates

The clean, guarded, `exec-run` application passed both pre-registered gates (`outputs/logs/*_apply_adjustment.log`):

- **Gate A:** median `|ret_adj|` across confirmed ex-days **0.508 → 0.037** (later 0.038 on the v3 panel’s 803 ex-days), with 99.3% of ex-day returns under 20% — corporate-action days became ordinary trading days.
- **Gate B:** extreme-day counts at the expected few-hundred-per-year scale, adjusted ≤ raw everywhere, largest survivors identifiable as suspension-relist artifacts (handled downstream by the ±50% clip), no split debris.

`returns_panel_v2.parquet` was thereby certified — and then superseded by one more identity-driven rebuild.

9. Identity: the ISIN closure problem

9.1 Why identity is a research problem, not a housekeeping detail

An ISIN is not a company. Splits and face-value changes mint new ISINs; mergers and demergers kill them; partly-paid and DVR lines coexist with the main line. Two failure modes threaten this project specifically: **fragmentation** (one company’s history split across ISINs — Alok Industries’ insolvency-and-relisting, a

quintessential HOSTAGE name, breaks into two unrelated stubs exactly at its distress event) and **false merging** (mapping a merger target onto its acquirer manufactures a fake continuous history). The v2 panel, keyed on raw ISINs, dropped 7.8% of model stock-days (pre-split history stranded under retired ISINs) and broke return chains at every face-value event.

9.2 First, the dog that didn't bark: the attrition diagnostic

Before building any closure, the 9.3% G2 join miss was decomposed read-only (`src/fii/validation/audits/module5d_attrition`) was it identity churn (fixable by a crosswalk) or genuine non-trading? The answer: **99.5% of unmatched stock-days were Cause B** — the cisin *is* in the tape but has no price that date (NSDL FII data is exchange-agnostic; BSE-primary days, pre-listing, post-delisting and suspension days produce an FII trade with no NSE close — correctly unmeasurable). A crosswalk would have recovered 0.05% of rows and was **not built** at that point. Equally important for inference: START- and END-anchor match rates were identical per archetype (HOSTAGE 95.9/95.9 in TEST), so reversal anchors were not differentially lost. The lesson worth generalizing: *diagnose attrition before engineering against it — most “missing data” problems are facts about the world, not defects in the pipeline.*

9.3 The closure rule (user-specified, audit-verified)

The eventual canonicalization — motivated not by the join miss but by the stranded pre-split history — is **issuer-bounded and corporate-action-type-conditional** (design record: `docs/research_log/FII_Module5_validation_log`; memory of the design debate in the project log):

1. **The entity key is inside the ISIN.** Characters 4–7 of an Indian ISIN are the issuer code (ONGC “213A” is stable even across an IN8→INE prefix change). Legitimate identity closure lives *within* one issuer code; any lookup link crossing issuer codes is a merger/acquisition candidate and must mean identity **death**, not mapping.
2. **Event type decides.** Value-preserving events (split, bonus, face-value change, rename) map old → the issuer's terminal (latest-trading) ISIN. Value-changing events (merger, demerger, amalgamation) terminate the old identity.
3. **Co-existence guard.** If two ISINs of one issuer *traded simultaneously for ≥ 180 days*, they are distinct lines (DVR, partly-paid) and are never collapsed.

The rule was not trusted on paper; it was audited against the data before being applied. `module5k_isin_accounting.py`: all 752 lookup links are same-issuer — zero cross-issuer links, so the entity-boundary condition already held; issuer-code chains decompose into 1,117 clean single-active chains (auto-closable), 168 fully-dead, and only 8 with >1 active line (the DVR/dual-class set — correctly not collapsed). `module5l_noisin_probe.py` closed the degenerate-key branch: fabricated `NOISIN<digits>` placeholders and null-ISIN records — a worry because they could either pollute the universe or fragment real companies — turned out to be 0.03% of raw trades, overwhelmingly *mutual-fund* records (non-equity, correctly out of scope), touching **zero** of the 946 model names. `module5m_universe_integrity.py` answered “is 946 even the right count”: 946 cisins = 939 distinct issuer codes; six of the seven doubled issuers were genuine fragmentation (Tata Steel, Alok, Ruchi/Patanjali, Bajaj Finance, Cholamandalam, Vaibhav Global), one a genuine co-existence (Bharti Airtel's partly-paid line). All audits are re-runnable: `python pipeline.py --phase audit`.

9.4 The v3 build and what it bought

`src/fii/data_prep/module5j_canonical_panel.py` (stage `canonical_panel`) applies the closure to **both sides** — the price tape *and* the model's state history (dual-side, so event windows and states agree about who

is who; day-labels themselves stay frozen). Outcomes (outputs/logs/*_canonical_panel.log):

- **9 model-side merges** (7 fragmented pairs + 2 terminal fixes): 946 cisins → **939 companies**.
- **Gate A re-passed on v3**: median $|\text{ret_adj}| = 0.038$ over **803** confirmed ex-days — 130 *more* verifiable ex-days than v2, because face-value splits that used to be null chain-breaks are now measurable across the canonical chain.
- **Coverage 90.4% → 98.49%**, uniform across archetypes (HOSTAGE 98.25%); TEST era 99.85%.
- **The 2011 backfill (R2) independently validated**: 98.45% coverage in 2011 — the recycling/timing risks flagged at repair time did not materialize.

Outputs: returns_panel_v3.parquet + states_v3.parquet (804,591 stock-days, 939 companies) — the certified basis for every result in Part IV.

Value at this stage, round five. Nothing in §§8–9 is glamorous, and all of it is load-bearing: the headline economic effect (~50 bp/20d) is three orders of magnitude smaller than one unadjusted split. The transferable assets: the Gate-0 pattern (*test the vendor's adjustment convention before using it* — NSE's prev_close is raw, contrary to common belief); the tape-verification loop for CA factors (parse → predict the ex-day ratio → confirm on the tape → null what fails); the application guard against symbol migrations; and the issuer-bounded, event-conditional, co-existence-guarded ISIN closure — to our knowledge not documented elsewhere for Indian data — plus the audit suite that certifies it. Any Indian-equities research group can lift Part III wholesale.

— *End of Part III. Part IV opens the economics: the event-study machinery and its inference correction (§10), the volume-signed mechanism (§11), the panel regression (§12), robustness (§13), the INNOV control (§14), the LightGBM challenger (§15), PIN (§16), the honest nulls (§17), and the backtests (§18).*

Part IV — Economic validation

Everything in Parts I–III was construction. This part is the confrontation with prices — designed as a gauntlet the hypothesis had to survive, and which, twice, it did not: the pre-registered story died and a different, better-supported one replaced it. Stages: event_study through bt_style_switch; every log in outputs/validation/.

10. Event studies: mathematics and the inference correction

10.1 The measurement object

The unit is the **episode** — a maximal run of consecutive same-archetype days in one stock. Two anchors, testing different economics: **START** (first day; tests drift while the behaviour unfolds) and **END** (last day; tests what happens *after the flow stops* — the correct anchor for any price-pressure story, because pressure relaxation begins when the pressure ends). For stock i with market-adjusted, CA-adjusted daily abnormal return $ar_{it} = r_{it}^{adj} - r_t^{NIFTY}$, clipped at $\pm 50\%$:

$$CAR_i(k) = \sum_{t=\tau_i+1}^{\tau_i+k} ar_{it}, \quad k \in \{1, 5, 10, 20\},$$

starting at $t + 1$, **never** at the anchor day itself: day-0's return is mechanically correlated with the flow that defined the label (same-day impact), so it is reported as context and excluded from evidence.

Three integrity provisions, each answering a specific bias (definitions: docs/research_log/FII_Module5_validation_log. §1):

- **Delisting truncation.** Events whose stock stops trading inside the window are truncated at the last price and **kept** — dropping them would delete exactly the fire-sale-to-bankruptcy outcomes the HOSTAGE test needs (survivorship bias where it hurts most). Truncation is slightly optimistic (post-delisting value ≈ 0), so `trunc%` is printed per archetype to keep the bias visible.
- **$\pm 50\%$ daily clip.** Kills suspension-relist artifacts (Rs.0.15 $\rightarrow$ Rs.8.55 prints +5600%) while leaving genuine circuit-limit days untouched.
- **Date-clustered bootstrap** ($N = 1000$). Events are not independent: one crisis day launches dozens of episodes that then share a market path. An i.i.d. t-test fakes precision by counting each as independent evidence. Instead, calendar anchor-days are resampled with replacement — each draw takes *all* events of that day — and the 95% interval and two-sided p come from the resampled distribution of mean CAR. The clustering unit (day) is the conservative choice.

10.2 The inference flaw we caught in our own first pass

The first model-facing run (`legacy/colab_modules/module5b3_car_start.py`; post-mortem in log §3c) tested CARs **against zero** — and everything was “significant,” including the placebo. Diagnosis: the TEST-era ALL_LABELED baseline — the mean forward CAR over *every* labeled stock-day — was **+52 bp/20d versus NIFTY**, a classic benchmark mismatch (beta-1 adjustment of mid-caps against a large-cap index in a 2021–25 mid-cap bull). Every archetype inherits that shift; only archetype — baseline is meaningful. The correction (`src/fii/validation/module5b4_car_diff.py`, stage `event_study`): report **excess CAR** with the date-clustered bootstrap run on the *difference*. This is a difference-in-differences: date-level shocks common to all archetypes cancel. The flawed vs-zero p-values were voided, and the episode is retained as a caution: *the most dangerous inference errors produce significance, not absurdity*.

10.3 The verdict that inverted the hypothesis (log §3f)

On the certified v3 panel, END-anchor excess CARs (`outputs/validation/event_study.log`):

Archetype	Prediction	TRAIN excess CAR20	TEST excess CAR20
HOSTAGE (dispersed sell)	reversal (the flagship)	−15 bp, $p=0.23$	+10 bp, $p=0.60$
SHARK_DIST (concentrated sell)	continued decline	+68 bp, $p=0.000$	+33 bp, $p=0.042$
SHARK_ACC (concentrated buy)	continued rise	−50 bp, $p=0.000$	−13 bp, ns
ROBOT (placebo)	≈ 0	≈ 0 , $p \gg 0.05$	≈ 0 , $p \gg 0.05$

The placebo behaving proves the method; the hypothesis rows falsify the hypothesis. The flagship HOSTAGE reversal is **absent** — null in both eras. And the “informed” Shark rows are *wrong-signed, significant, and replicating*: concentrated selling is followed by price **recovery**; concentrated buying by **give-back**. The data-driven reframing: **the concentration axis marks temporary price pressure** — the Coval–Stafford reversal mechanism is real, but it lives on the *concentrated* episodes, not the dispersed ones. This was inversion number one. It demanded a mechanism check before being believed.

11. The mechanism: a liquidity shock signed by volume

11.1 The corroboration that failed, informatively (log §3g)

If concentrated FII selling is block-like liquidity demand, it should coincide with exchange-disclosed block/bulk deals. Tested (`src/fii/validation/module6_deal_corroboration.py`): SHARK_DIST days carry *fewer* sell-side deal disclosures than HOSTAGE days (0.73% vs 0.96%, ratio 0.76×, $z = -4.2$) — opposite to prediction — and within every archetype, buy-deal and sell-deal rates are nearly equal, revealing that disclosed-deal incidence is a two-sided *activity/attention* proxy, not a directional-pressure measure. Three structural confounds explain the null: disclosure thresholds (0.5% of shares, single orders) miss institutional orders *worked* across a day; the disclosure universe is all participants (promoters, domestic funds) while our concentration is FII-only; and HOSTAGE days are simply eventful. At this point two economic interpretations were dead (informed/forced from §10; block-visibility from here), and a decision rule was imposed to prevent interpretation-shopping: **one final mechanism test with a pre-committed stopping rule** — if it failed, the reversal would be published as an unexplained regularity, full stop.

11.2 The pre-committed test, and its confirmation (log §3h)

If concentrated episodes are liquidity shocks, they must carry the *unmaskable* signature: price pressure during the episode **and elevated trading volume at the climax**, followed by recovery. The bar, written before the run: SHARK_DIST must show episode $CAR < 0$ *and* day-0 relative volume > 1 *in both eras*, or mechanism-hunting stops. The full arc (`src/fii/validation/module6b_liquidity_shock_profile.py`; figure outputs/figures/F1_mechanism_arc.png, table outputs/tables/T3_mechanism_arc.csv):

- **SHARK_DIST**: $\approx -270/-226$ bp excess decline in the month *before* the episode $\rightarrow$ episode-leg pressure (epCAR $-17/-5$ bp, medians $-18/-12$) with **day-0 volume above baseline** $\rightarrow +81/+32$ bp excess recovery after the END — roughly one-third of the preceding decline retraced.
- **SHARK_ACC mirrors with flipped signs**: $+305/+359$ bp run-up, $+70/+48$ bp episode pressure, elevated volume, $-49/-15$ bp give-back.
- **HOSTAGE — the decisive contrast**: *no* episode pressure, volume **below** baseline (0.95–0.98×), and no recovery, ever. Dispersed selling is *quiet* distribution the market absorbs without a liquidity premium — and its price marks are permanent.
- **ROBOT**: flat arc, $\approx 1.0\times$ volume. The volume ordering SHARK_DIST $>$ SHARK_ACC $>$ baseline $>$ HOSTAGE replicates across eras.

The pre-committed bar was met in both eras. The final economic reading — the sentence the whole project compresses to — became:

The concentration of FII participation separates transitory from permanent price impact.
Concentrated flow is liquidity-demanding: volume-marked, price-pressuring, partially reverting.
Dispersed flow is information-consistent: quiet, and permanent. The archetype names inverted;
the mechanism is coherent, replicating, placebo-clean, and established from internal data alone.

It also retro-explains the block-deal null: disclosure data measures a *visibility threshold*; volume — which cannot be masked — carries the actual footprint.

12. Panel regression: fixed effects, clustering, and the bad control

12.1 Why a regression after event studies

Event-study differences control for *era-level* drift but not for the possibility that archetype episodes happen to sit on particular *stocks* (cheap, illiquid, high-beta) or particular *dates*, or that “reversal

after concentrated selling” is just the generic reversal of any recent loser. The formal machinery (src/fii/validation/module7_panel_regression.py, stage panel_regression; estimator: linearmodels.PanelOLS — a deliberate choice after a user correction that statistical models must be *visible library calls*, auditable against documentation, not hand-rolled algebra):

$$y_{ie} = \alpha_i + \delta_{t(e)} + \sum_a \beta_a D_{ie}^a + \gamma' \mathbf{z}_{ie} + \varepsilon_{ie},$$

where e indexes episode-END events, y is the forward 20-day abnormal return (bp), D^a are archetype dummies (omitted category: UNTAGGED — the “directional but unremarkable” days, the natural control), α_i are **stock fixed effects** (absorbing every time-invariant stock attribute: sector, average liquidity, governance, listing venue), and δ_t are **date fixed effects** (absorbing every market-wide date shock: index moves, macro news, VIX level — which is *why* VIX appears nowhere until §17: it is implicitly controlled here). Estimation is the within transformation — demeaning y and regressors by entity and time — so β_a is identified purely from *within-stock*, *within-date* variation: “on the same date, did the stock whose concentrated-sell episode just ended outperform another stock of its own average behaviour?”

12.2 Inference: two-way clustered standard errors

Residuals correlate both within a stock over time (persistent stock-level shocks) and across stocks within a period (common shocks the date FE don’t fully absorb, plus the mechanical overlap of 20-day windows). Standard errors are therefore **clustered two ways, by stock and by calendar month**, via the cluster-robust sandwich estimator $\widehat{V} = (X'X)^{-1}(\sum_g X'_g \hat{u}_g \hat{u}'_g X_g)(X'X)^{-1}$ summed over clusters g , combined across the two dimensions by inclusion–exclusion (Cameron–Gelbach–Miller). Clustering by *month* rather than day for the time dimension is the conservative choice given 20-day overlapping outcomes.

12.3 The specification ladder and the bad control, reasoned

- **R0:** fixed effects only.
- **R1:** + characteristics measured strictly before the anchor: rolling 120-day beta, momentum (t–126...–21), Amihud illiquidity, log turnover, 20-day volatility, relative volume, log price, log episode length.
- **R2:** R1 + **the pre-episode 20-day return (pre20)** — included knowing it is a *bad control* in the causal sense: the episode itself produces price pressure (§11), so pre20 is partly a **mediator**, and controlling for it absorbs part of the true effect. That is the point: R2 is a deliberately conservative lower bound that mechanically soaks up any “losers bounce” channel. If the archetype dummy survives R2, the generic loser-reversal objection is dead by construction. R1-vs-R2 brackets the effect.

One econometric note we state rather than bury: archetype labels are **estimated regressors** (outputs of a fitted model). Classification error in a dummy attenuates its coefficient toward zero — so reported effects are, if anything, understated.

12.4 Results and the two flags (logs §3i–3j; outputs/tables/T1_panel_regression_main.csv, full coefficients outputs/regression_outputs/)

The headline (R2, restored full sample — see flag 2):

END-event dummy	TRAIN	TEST (OOS)
SHARK_DIST	+65.4*, t=5.35	+48.6*, t≈2.9

END-event dummy	TRAIN	TEST (OOS)
SHARK_ACC	−87.9*, t=−6.23	−47.6*, t=−2.84
HOSTAGE	−3.6, ns	−8.1, ns
ROBOT	−7.7, ns	−38.4*** (see §13)

with $n = 63,259 / 35,572$ episode-ends. (Bounce-free companions, windows $t+3..t+22$: SHARK_DIST +58.0*** / +36.7** — the first-two-day bid-ask component is 11%/24% of these coefficients; measured in §13.5b and quoted alongside the headline wherever it matters.) Controls behave sensibly (log price strongly negative — the microcap-bounce effect; Amihud positive in TEST — an illiquidity premium), and pre20 enters significantly in TRAIN while the SHARK_DIST dummy *barely moves* from R1 to R2 — generic loser-reversal exists in this market and the archetype effect is not it. Two honest flags were raised in the first pass and resolved in §13: the TEST-era ROBOT placebo failure, and a TRAIN sample collapse (66k→27k) in R1/R2 caused by sparse index-return nulls poisoning 120-day rolling betas (one null kills a window; zero-filling 0.6% of index gaps restored n with coefficients essentially unchanged — proof the collapse was never selection).

13. Robustness: overlap, horizons, dose–response, and the placebo that wasn’t

All in `src/fii/validation/module7b_robustness.py` (stage `robustness`, log outputs/validation/robustness.log).

(A) The ROBOT “placebo failure” is structural, and the honest fix is a better control. Decomposing ROBOT-END events by *what follows them*: a ROBOT (neutral) episode ends **by transitioning into a directional regime**, so its post-window mechanically contains the successor regime’s flow. ROBOT post20 splits +94 bp after BUY-transitions vs −13 bp after SELL-transitions; the TEST-era negative aggregate is transition mechanics plus delisting truncation (327 truncated events at −129 bp), not a leak in the machinery. Conclusion: ROBOT-END is structurally invalid as a placebo *at the END anchor*. The valid null is **HOSTAGE** — a selling episode differing from SHARK_DIST only in concentration — which is clean everywhere. We report this rather than hide it: a reader should see that one pre-registered control failed *for a reason the data explains*, and that the mechanism-relevant control is stronger than the failed one.

(B) Non-overlapping episodes. Within stock, greedily keep episodes ≥ 28 calendar days apart (39.6k of 102.9k kept): SHARK_DIST **strengthens** (+76/+108). The overlap objection — that clustered episodes double-count one price path — is dead; if anything, overlap *diluted* the effect.

(C) Horizons. R2 dummies at 10/20/30/60 days (outputs/tables/T4_horizons.csv, figure F2_horizons.png): SHARK_DIST builds +39 → +65 → +89 → +109 bp (TRAIN) and +24 → +49 → +51 → +73 (TEST); HOSTAGE is ≈ 0 at every horizon. A transitory-impact signature must build and persist — a momentum artifact would decay, a microstructure bounce would flip. This panel is the single strongest exhibit in the project.

(D) Continuous dose–response. Replace dummies entirely: within sell-regime episodes, regress post20 on the episode-mean concentration $\bar{F}_e^{entity_s}$ (no thresholds, no labels — sidestepping estimated-regressor attenuation). All four era×side cells are right-signed; TRAIN-SELL is cleanly significant (+27.2 bp per unit, $p < 0.01$); TEST-SELL is +15 bp, $p = 0.16$. Read honestly: the *linear* slope is diluted where the *tail* dummy is strong — the reversal is a tail-of-concentration phenomenon, consistent with block-like liquidity demand being an extreme event, and TEST is underpowered for the continuum version. We claim the tail effect, not a universal linear law.

13.5 The referee tests (Module 13)

An external review conceded the methodology and challenged the economics on three testable grounds. Each became a pre-registered experiment (scripts module13a/b/c, stages backbone_ablation, incremental_value, recon_exclusion; log §3p):

(i) **“Is the HMM central or decoration?”** Answered by the ablation in §5.6: a census-matched persistence rule reproduces Table 1 within band ($\kappa = 0.891$ label agreement). Verdict V1 — the HMM is not necessary; the contribution is the measure. Adopted as the paper’s framing.

(ii) **“Show the measure adds information beyond existing predictors.”** Two instruments. At the *episode* level, adding the obvious competitors — episode-mean flow imbalance, log flow size, scaled net flow — to the R2 spec moves D_SHARK_DIST by only $-13\%/-11\%$ ($+65.4 \rightarrow +57.1^{***}$ TRAIN, $+48.5 \rightarrow +43.4^{***}$ TEST): **composition is not flow magnitude in disguise**. At the *daily* level, a LightGBM fitted with versus without the composition block ($\{F_entity, F_entity_buy, F_breadth\}$; 788,935 stock-days) gains $\Delta IC = +0.0060$ with paired daily $t = +3.09$ — a **28% relative improvement** ($0.0211 \rightarrow 0.0272$) that cleared both pre-registered numeric bars — while the *quintile spread* did not improve, and the composition-only model showed the inverse profile: the lowest IC ($+0.0043$) but the largest and only significant spread ($+73.2$ bp, non-overlap $t = 2.84$). Read together: composition ranks the broad cross-section weakly but its extremes are potent — a third independent confirmation (with the dose-response and the dummy-vs-continuous contrast) that the signal’s information is **tail-concentrated, exactly where the paper claims it**.

(iii) **“Could it be ETF/index rebalancing?”** Excluding every episode ending within ± 7 calendar days of any of 150 MSCI/FTSE/NIFTY review dates — discarding 41% of episodes — leaves Table 1 intact (SHARK_DIST $+61.5^{***}$ TRAIN / $+50.9^{**}$ TEST; at ± 3 days, $+72.2/+59.1$, if anything stronger). **Passive index mechanics are excluded** as the driver.

The boundary we concede rather than test. The reviewer’s sharpest sentence — “*your variable measures how trading is distributed, but the interpretation extends to why trading occurred*” — is correct as a boundary statement, and this document’s language is drawn at exactly that line. What is measured: transitory versus permanent price impact, sorted by participation concentration. What is supported but not proven: the information interpretation of the permanent side (“information- *consistent*”, backed by the count-based PIN loading, §16). What cannot be distinguished with this data: informed dispersed selling versus non-informational permanent-impact channels such as ESG or mandate-driven exits under downward-sloping demand — both produce quiet, permanent declines, and stock-level recon membership is likewise untestable with a market-wide date file. These live in §19’s ledger and the limitations, not in the claims.

13.5b The skeptic tests (Module 15): four holes we found in our own work, two claims repriced

After every external round closed, a deliberately hostile re-read of the research (not the paper) surfaced four first-order objections **no reviewer had raised**: (i) bid-ask bounce — episode ends after concentrated selling close at the bid, so mid-reversion mechanically fakes a “+reversal”; (ii) the nearest prior art is not the block-trade literature but volume-conditioned reversal (Conrad–Hameed–Niden 1994), never confronted head-to-head; (iii) the paper’s headline contrast (“SD reverts, HOSTAGE doesn’t”) was supported by “significant vs not significant,” which is not itself a test; (iv) the SUB_ACC robustness promised in Module 1 was never executed. The first three were tested (module15_skeptic_tests.py, stage skeptic_tests, pre-registered; log §3r); the fourth is stated as an open item.

Results, with the honest repricing. (i) Bounce-free windows ($t+3..t+22$): SHARK_DIST $+58.0^{***}$ / $+36.7^*$ vs the standard $+65.4/+48.5$ — a bounce component of **11% (train) and 24% (test)**. Bounce is not the driver (the effect survives and continues building), but headline magnitudes are henceforth quoted with

their bounce-free companions, and the execution-savings translation scales down accordingly. Interpretive note: the spread concession *is* the liquidity-demand mechanism in miniature, so this component is part of the story, not an artifact of it — it is simply not novel. (ii) A public volume-conditioned-reversal dummy (bottom-quintile loser with $\text{relvol} > 1.1$ — computable from free data) enters weak and insignificant (+26/+21 bp) while D_SHARK_DIST is unchanged to the first decimal beside it; and the GBT ladder gives the definitive incremental-value hierarchy: PUBLIC 0.0160 $\rightarrow$ +composition 0.0226 (paired $t = 5.20$); PUBLIC+conventional-flow 0.0289 $\rightarrow$ +composition 0.0330 ($t = 3.23$). **Composition adds significant information beyond free public data and beyond conventional private flow.** (iii) The direct contrast $\beta_{SD} - \beta_{HO}$ is significant in the standard window in both eras (+69.0, $t = 4.49$; +56.6, $p = 0.016$) — but in the bounce-free test-era window it is +30.1 with $t = 1.31$: **directionally right, underpowered.** The sharpest version of the headline sentence is therefore stated with that qualification wherever it appears, and the permanence claim carries its power bound (a HOSTAGE reversal smaller than ~ 40 bp would be undetectable here).

13.6 The tail probe (Module 14): the reversal is a liquid-market phenomenon

A second external critique argued the opposite of caution: that the *real* mispricing lives in the illiquid 75% we excluded, where dislocations are hundreds of basis points. This was testable — and the 13A rule backbone made it feasible (no HMM, no sequence floors). Three pre-registered steps (module14a/b/c; log §3q): a feasibility census (the *measurable* tail is the attributable tail — 137,774 stock-days across 1,618 names; gates passed), a method-frozen label build (backbone cuts re-derived on tail-TRAIN landed at $-0.545/+0.556$, strikingly near the liquid universe’s; episode clustering $2.25\times/2.86\times$, $p = 0.005$ — the temporal structure exists out there at full strength; one implementation failure, #12 in the ledger, caught by its own impossible printout), and the economics.

The verdict, pre-registered: NO TAIL EFFECT. Tail concentrated-selling episodes show *bigger* dislocations (pre-episode $-455/-225$ bp; episode leg -196 bp) and a *stronger* volume climax (day-0 volume $3.42\times/1.58\times$ trailing) — the mechanism’s fingerprint transfers emphatically. The recovery does not: post-END excess CAR20 is +3 bp (TRAIN, $p = 0.93$) and +36 bp (TEST, CI $[-51,+128]$, $p = 0.37$), with even the optimistic point estimate at one-third of the tail’s own friction bar (Roll round-trip spread $\sim 98-115$ bp, against 156 bp median *daily* moves).

The economic reading upgrades the paper’s story: **the post-episode reversal is itself liquidity provision** — it exists only where arbitrage capital operates. In the liquid 946, someone steps in within a month and corrects the overshoot; in the tail, the dent stays. Transitory price impact is a liquid-market phenomenon, and limits-to-arbitrage binds at *both* ends of the spectrum: in liquid names the signal is smaller than the costs; in illiquid names the reversal does not exist and the costs triple. The long-standing scope caveat (“fire-sale effects may be *stronger* in the excluded tail; our estimates are a floor”) is hereby **revised by measurement**: for the reversal side it was wrong in the hedged direction. One suggestive cell is reported without being claimed: tail HOSTAGE episodes in the test era show *continuation* (-95 bp, CI $[-167,-20]$) — dispersed selling in illiquid names keeps falling — directionally consistent with the information story, but a single significant cell in a coverage-confounded era.

14. Alternative explanations and the flow-surprise control (INNOV)

The remaining alternative after §§12–13: maybe “concentration” is just **flow-surprise magnitude** in disguise — big unexpected flow both looks concentrated and mean-reverts. Closing this required a flow-expectation model (src/fii/validation/module9_net_innov.py, stage flow_innovation).

Construction. Per stock, an AR(5) on scaled net flow $n_{it} = NET_{it}/\overline{GROSS}_i$:

$$n_{it} = c_i + \sum_{k=1}^5 \phi_{ik} n_{i,t-k} + u_{it}, \quad INNOV_{it} \equiv \hat{u}_{it},$$

fitted per stock by least squares over the **full sample — with deliberate look-ahead**, and the rationale is on record because it was demanded (log §31): (1) INNOV is a *yardstick*, not a claim — full-sample coefficients give the stablest expectation and the cleanest decomposition; an expanding-window version injects estimation noise into a control; (2) the bias direction is **conservative for us** — INNOV is the competing explanation, and hindsight makes it the best-case competitor, so the archetype surviving against an information-advantaged control is the *stronger* result; (3) the asymmetry is the design: the archetypes (the claim) are strictly real-time; INNOV (the instrument) is allowed hindsight. Sanity: INNOV correlates with same-day returns at IC +0.33/+0.28 (99–100% of days positive) — it measures what it should.

The key test (T3). Add episode-mean INNOV to the R2 regression: **D_SHARK_DIST is unchanged to the decimal in TRAIN (+65.4) and remains +43.6** in TEST, while mean-INNOV itself enters at –25/–32*** — surprise-magnitude reversion is real *and separate*. Concentration is not flow-surprise in disguise. A non-parametric backstop (T4): SHARK_DIST out-reverts HOSTAGE within every TRAIN INNOV-tercile and 2 of 3 TEST terciles (one exact tie, reported).

The bonus regularity (T2), with its caveat welded on. INNOV predicts *negative* forward returns in both eras (daily IC –0.023/–0.017, $t = -3.2/-2.6$): unexpected FII flow reverts. Because the expectation model is in-sample, this is a **decomposition result** — FII flow contains a transitory component whose price effect reverts — and *not* an implementable signal. It is the project’s second regularity, always stated with that caveat.

15. The machine-learning challenger: why LightGBM won the battle and lost the war

15.1 The question the challenger answers

Is the HMM the *bottleneck*? The regime pipeline uses four features and a particular model class; perhaps a flexible learner on all ten features finds forward-return structure the regimes miss — in which case the right next step is better machine learning, not more economics. The design (src/fii/validation/module8_gbt_shap.py, stage gbt_challenger) pits LightGBM against a **regime baseline** (each archetype’s TRAIN-mean outcome as the prediction — the information content of the labels alone), with **pre-registered success bars**: test-era IC gap > 0.01 over the baseline *and* top-minus-bottom quintile spread with non-overlap $t > 2$.

15.2 The mathematics being deployed

Gradient boosting fits an additive model $\hat{y}^{(M)}(\mathbf{x}) = \sum_{m=1}^M f_m(\mathbf{x})$, each f_m a regression tree chosen to minimize the second-order Taylor approximation of the loss around the current prediction: with $g_i = \partial_{\hat{y}} \ell(y_i, \hat{y}_i)$ and $h_i = \partial_{\hat{y}}^2 \ell(y_i, \hat{y}_i)$, the optimal leaf weight is $w_L^* = -\sum_{i \in L} g_i / (\sum_{i \in L} h_i + \lambda)$ and the split gain is the induced reduction in $-\frac{1}{2} \sum_L (\sum_{i \in L} g_i)^2 / (\sum_{i \in L} h_i + \lambda)$. LightGBM grows trees **leaf-wise** (always splitting the highest-gain leaf) on **histogram-binned** features — fast, and natively tolerant of missing values, which matters because the feature store’s nulls are honest (§3.5) and must not be imputed. Configuration: 400 trees, learning rate 0.05, 63 leaves; target: forward 20-day abnormal CAR; same universe, same frozen split.

SHAP attribution. Shapley values from cooperative game theory: feature j ’s contribution is its marginal effect averaged over all orderings of feature inclusion, $\phi_j = \sum_{S \subseteq F, j \notin S} \frac{|S|!(|F|-|S|-1)!}{|F|!} [v(S \cup \{j\}) - v(S)]$ — the unique attribution satisfying local accuracy, consistency and symmetry. For trees this is computed *exactly* (TreeSHAP / pred_contrib=True), so the attributions are not approximations.

15.3 What happened — including a bug we own

The challenger *beat the baseline*: TEST IC +0.0279 vs +0.0117 (gap +0.016 > 0.01 bar), monotone quintiles, Q5–Q1 = +74.5 bp/20d with non-overlap $t = 3.22$ (> 2 bar); TRAIN IC 0.218 — heavy memorization, expected of 400 trees, which is precisely why only TEST counts. SHAP said the edge was *not* mainly the concentration tail: top features were F_entity_buy (ride the pressure while flow persists — an *arc-timing* signal a day-label cannot express), the contrarian F_imbal/F_streak (features the HMM never used), and F_breadth. One artifact was caught and is owned: the module’s single-feature IC table lacked a per-day dropna, producing a spurious –0.135 “IC” for F_breadth (recomputed correctly: +0.012). The GBT itself was unaffected (LightGBM handles NaN natively); the correction is recorded in log §3k.

15.4 The decomposition that settles it (module8b_demeaning_check.py)

Before concluding “sequence models next,” a gate: how much of the GBT edge is *dynamic* (which-day information) versus *cross-sectional* (which-stock information — characteristics in disguise)? Method: within-stock demeaning using **TRAIN-era stock means** (test-safe), then (a) recompute feature ICs — essentially unchanged, so no feature’s signal is purely a stock identity — and (b) refit the GBT on demeaned features, isolating dynamics. Result: dynamics-only IC +0.0161, spread +53.4 bp, monotone — but **non-overlap $t = 1.48 < 2$: the pre-registered bar for the dynamic increment is not met**. Also measured: F_breadth is the most characteristic-like feature (55% between-stock variance, 0.77 TRAIN→TEST rank stability), explaining part of the full-GBT edge as stock selection.

So why did LightGBM not deliver a big advantage? Because its edge decomposes into (i) the regime information itself, (ii) static stock characteristics that fixed effects already handle in the economics, and (iii) a *sub-significance* timing component. Held to the pre-commitment: **the HMM is not the bottleneck; the features are the ceiling at this data’s statistical power; LSTMs/sequence models are not justified**. The one identified lever for future work — the top dynamic SHAP feature — is within-stock *breadth change* (FII participation surges), which the current regime rules never use.

16. External validation: the PIN model

16.1 Why PIN, and why it is genuinely independent

Every test so far shares one input: returns. The Easley–O’Hara PIN (probability of informed trading) is estimated from **order counts only** — it has never seen a price, a return, or an archetype definition — so its agreement is external corroboration in the strict sense. Our variant uses FII buy/sell trade counts per stock-day (“FII-PIN”; stated limitation: classical PIN uses all market orders).

16.2 The model and its estimation (src/fii/validation/module11_pin.py)

Each day is no-news (prob $1 - \alpha$), bad-news ($\alpha\delta$) or good-news ($\alpha(1 - \delta)$); buys and sells arrive Poisson with uninformed rates $\varepsilon_b, \varepsilon_s$ and an informed rate μ added to the news side. The day’s likelihood is the three-component mixture; the stock-year log-likelihood is maximized by L-BFGS-B with Lin–Ke stabilization (log-sum-exp over components; the classical EHO likelihood overflows on high-count days) and logit/log parameter transforms. Then

$$PIN = \frac{\alpha\mu}{\alpha\mu + \varepsilon_b + \varepsilon_s},$$

the share of order flow attributable to informed trading. Estimated per stock-year (≥ 60 days), 4,456 stock-years, 826 stocks; output data/VALIDATION_DATA/fii_pin_stockyear.parquet.

Sanity before use: PIN levels median 0.19, IQR 0.14–0.25 (the literature band), and correlation with log turnover -0.117 — informed trading probability higher in smaller names, the canonical sign.

16.3 The result (T2; outputs/tables/T5_pin_loadings.csv, figure F4)

Regressing stock-year PIN on the year’s archetype shares (turnover controlled, stock-clustered SE):

	TRAIN	TEST
HOSTAGE share	+0.231* , t=7.9	+0.149* , t=4.5
SHARK_DIST share	+0.077***, t=2.9	+0.065**, t=2.5

Dispersed-selling exposure carries $\sim 3\times$ **the informed-trading loading** of concentrated-selling exposure, in both eras — an independent construct arriving at the same transitory/permanent assignment as the price tests. Two disciplined nuances: the pre-registration said SHARK_DIST ≈ 0 and it is small-but-significant, so the claim is **relative** (concentrated selling carries *less* information, not none); and T3 confirms the liquidity reading from the other side — the SHARK_DIST reversal is *not* concentrated in high-PIN names (it is larger where PIN is low), exactly as a liquidity effect should be, while HOSTAGE is ≈ 0 in both PIN halves (permanence is not hiding a conditional bounce).

17. State dependence: the honest nulls (VIX, Kyle’s λ)

Late in the project the user noticed India VIX had “never been used.” The first half of the answer is econometric: **VIX levels were controlled all along** — date fixed effects absorb every market-wide variable, VIX included. What FE cannot absorb is an *interaction*: is the reversal bigger in high-VIX states? Tested (src/fii/validation/module10_vix_lambda.py): D_SHARK_DIST $\times$ high-VIX is $+83^{***}$ in TEST — the entire TEST reversal living in high-VIX halves, a textbook liquidity signature — but -27 (ns) in TRAIN. By this project’s replication standard, **non-replicating** $\rightarrow$ **suggestive, not claimable**, and that is how it is reported.

The second candidate: a FII-flow **Kyle λ** per stock — trailing 120 flow-day $\text{cov}(ar_t, \tilde{f}_t)/\text{var}(\tilde{f}_t)$ with scaled net flow $\tilde{f}$, past-only (median $+0.0043$, sane) — does the reversal amplify in high- λ names? The interaction is **null in both eras**. Raw terciles *do* show $2.5\times$ amplification in high- λ names, but the main effect exposes it: high- λ names bounce more after *all* episode types — an illiquidity characteristic, not an archetype mechanism — and the FE machinery is what prevented that false claim from being made. Both nulls are reported at full length precisely because a validation battery that only ever confirms is not a validation battery.

18. Economic significance: backtests and limits to arbitrage

18.1 Why backtest a research finding at all

Statistical significance at 50 bp/20d says nothing about economic harvestability. The user’s question — build strategies with and without the model and compare — is the correct final test, and it was run under the same discipline as everything else: a gated engine, frozen baselines, pre-registered verdicts (src/fii/backtest/; full detail docs/05_backtesting.md; log §30).

The engine (engine.py) is a daily cross-sectional long/short machine: $\alpha \rightarrow$ within-day demean $\rightarrow$ unit gross $\rightarrow$ 5% position cap; signals formed at close t trade at close $t+1$ and earn day $t+2$ (one-day lag); costs 15 bps one-way on turnover. Its own correctness is gated (stage engine_gates): vectorized-equals-naive-loop exactness; a **cheat-alpha alignment test** (tomorrow’s return as today’s signal: Sharpe $+211$ at lag 0, collapsing

to -0.5 at lag 1 — proving both the alignment and that the lag genuinely delays); and exact cost/turnover accounting on a constructed flip book. Episode ENDS are detected real-time-legally: an end is knowable only at the close of the first day *after* the run.

18.2 Strategy pairs and verdicts (tables T6_backtest_metrics.csv, figure F3)

Four pairs, each a no-model baseline (frozen first) against an HMM-conditioned twin; verdict rule: TEST-era net-of-cost paired Δ Sharpe with a 20-day-block bootstrap 95% CI. Headline numbers, TEST era:

Book	Gross Sharpe	Breakeven cost (bps, one-way)	Net Sharpe @15 bps
S3 mechanical concentration proxies	+1.44	7.2	-1.54
S2H flow-following ex-concentrated-days	+1.46	4.3	-3.62
S4H style-switch on regimes	+0.54	1.7	-4.08
(every book)	—	2–8	negative

Three findings, in decreasing order of confidence. **First**, the model’s information is real at the portfolio level and appears exactly where the research says it should: removing concentrated-regime days improves a flow-following book’s *gross* Sharpe in **both eras** (Δ Sharpe CI [+0.03,+0.22] train, [+0.07,+0.49] test — the pre-registered “signal real, costs bind” verdict), and the regime-arc style-switch book (trend within concentrated episodes, reversion after their ends, flat elsewhere) is the only strategy gross-positive in both eras while its mechanical twin is gross-negative. **Second**, honest verdict accounting: of the formal net-of-cost verdicts, one strategy was inconclusive, two read “HMM subtracts” (one of them, S3H, had a construction flaw — unequal leg magnitudes made it effectively a HOSTAGE-short churn book — owned in the log), and S4H’s formal “HMM adds value” is driven mostly by its twin’s cost bleed; the defensible claim is the gross one. **Third**, the wall: breakeven costs of 2–8 bps one-way against realistic 15 bps mean **nothing here is a trading strategy at daily rebalance** — the reversal is ~ 50 bp/20d on $\sim 7\%$ of stock-days.

18.3 The limits-to-arbitrage reading

This is not a disappointment to be minimized; it is the *explanation of the regularity’s persistence*. An anomaly that survives fourteen years in a heavily-arbitraged market must have a reason arbitrageurs leave it alone: harvesting it costs more than it pays. That also identifies who *can* use the signal — anyone already obligated to trade (execution desks scheduling around concentrated-flow episodes; PMs deciding whether a decline is §11’s quiet-permanent kind or the volume-marked-reverting kind), for whom the transaction costs are already sunk. Short legs face India’s securities-lending constraints on top — stated, as everything else, rather than assumed away.

— *End of Part IV. Part V closes the ledger: every assumption in one table, the value audit by audience, the complete failure log, and the conclusion.*

Part VI — Phase II: from validated states to prediction

23. Causal estimation, calibrated forecasting, and the decision layer

Phase I established the states as measurement; Phase II (charter: docs/PHASE2_PLAN.md, pre-registered; log FII_Phase2_log.md) asked the engineering question: *using only information through t, estimate the current regime, forecast its evolution, and convert forecasts into decisions — under walk-forward discipline*. Data horizon: the sample ends 2025-03-28; all folds and claims respect it. Four stages, four verdicts, two of them failures worth their weight.

16A — the causal foundation (PASS). Phase-I labels were smoothed (full-sequence Viterbi: the state at t sees the future). The filtered posterior $P(S_t | x_{1:t})$ from the frozen parameters recognizes new regimes with a median lag of one day (p90: 3), agrees with the smoothed labels 91.3%, and — the gate — Table 1 on fully causal labels is intact (SHARK_DIST +60.2/+50.3; test marginally *stronger*). With 13A, this closes the look-ahead question from two independent directions.

16B — calibration (original gate FAIL, on the record; amended gate PASS). Our first persistence baseline conditioned on yesterday's *smoothed* label — unknowable at time t and quasi-circular with the smoothed target; the filtered nowcast lost to it on Brier (won on log-loss) and the pre-registered gate failed as written. The amendment (documented before rerunning) replaced it with the information-set-fair baseline — persistence of yesterday's *filtered* state — which the nowcast beats decisively (Brier 0.121 vs 0.262; log-loss 0.182 vs 0.532). Calibration is excellent where 79% of the probability mass lives (conf 0.99 → hit 0.994) and overconfident mid-range (0.70 → 0.55) — a measured input caveat for anything consuming these posteriors.

16C — the hazard model (PASS, decisively). The economically relevant forecast is the episode END (the reversal starts when concentrated flow stops). A walk-forward gradient-boosted discrete hazard (yearly re-fits 2014–2025Q1; 14,273 causal episodes; right-censoring handled; runs broken on >21-day calendar gaps) forecasts “today is the last day” at AUC **0.797 against the age-only baseline's 0.569** — out-of-sample era *stronger* than in-sample (0.809 vs 0.786) — with log-loss improvements at paired $t = +42$ ($k=1$) and $+34$ ($k=3$). Episode ends are genuinely forecastable beyond duration alone.

16D — the decision layer (FAIL on its pre-registered bar — the instructive result). Entering on the model's end-forecast instead of waiting for confirmation adds **+23 bp/episode out-of-sample (CI [+7,+40]), cost-neutral** (same single entry, earlier). But an age-only anticipator — “enter once the episode is a few days old,” three lines, no model — earns +18 bp, and paired on common episodes the model is significantly *worse* (−15 bp, $t = -2.96$). Diagnostics: the threshold selection was fold-unstable, and roughly half the anticipation gain is the first post-end day, which §13.5b already identified as partly spread concession.

Phase II conclusion. The framework works end-to-end — causal estimation, calibrated probabilities, real forecast skill — and its decision value is capped by exactly the ceilings Phase I measured (8B's sub-significant dynamic increment; Module 12's cost wall). The honest sentence: *episode ends are forecastable but not monetizable beyond a trivial age rule; the practical ~20 bp/episode of anticipation value requires no machine learning at all*. For the execution-overlay story, that simplifies the advice: do not wait for confirmed episode ends — act once a concentrated episode is a few days old. The hazard model earns its keep as monitoring (calibrated end-probabilities), not as timing alpha. Both failed gates in this phase are reported as failed; both taught more than the passes.

Part V — The ledger

19. Assumptions, in one place

Every result in Parts II–IV rests on assumptions. Scattering them across sections is how papers hide fragility; here they are collected, each with its status: **tested** (a check exists in the pipeline), **bounded** (the direction of the induced bias is known and stated), or **maintained** (assumed, with the argument for why).

#	Assumption	Where it bites	Status
A1	Masked entity IDs are internally consistent <i>within</i> a month	Axis 3 / all concentration measures	Tested — per-month counts sane (~150 brokers, ~3,000 FIIs); cross-month identity separately <i>disproved</i> (§2.2) and never used
A2	Value-weighted ID coverage $\geq 50\%$ $\rightarrow$ concentration measurable	late-sample (2024–25) HOSTAGE rows	Tested + bounded — coverage holds value-weighted; late-era HOSTAGE <i>prevalence</i> still flagged uninterpretable (log §4, Module 3)
A3	Within-day rank-probit preserves the economically relevant signal	every feature	Maintained + bounded — buys stationarity; forfeits market-wide level information by construction (census shares pinned)
A4	First-order Markov states, diagonal-Gaussian emissions	HMM backbone	Tested indirectly — OOS replication of signatures/census/transitions (§7.2); diagonal justified by near-orthogonal features (§4.2)
A5	20-day windows / 5-day HHI smoothing are the right time scales	features, overlays	Maintained — dwell times (13–17d) and episode medians (2–3d) are outputs, not inputs, and came out economically sensible; window sweep not exhaustively run
A6	Train $\leq 2021-04$ generalizes to test $\geq 2021-07$	everything frozen	Tested — the entire Part IV replication structure

#	Assumption	Where it bites	Status
A7	NSE bhavcopy close-to-close per canonical ISIN = investor return	all CARs	Tested — Gate A (0.508→0.038); bounded residuals: dividends unadjusted (small at daily scale), rights unadjusted (333 events, counted), relist artifacts clipped at ±50%
A8	Beta-1 market adjustment (NIFTY subtraction)	CARs, arc	Bounded — the +52bp TEST drift it creates is neutralized by baselines/date FE; rolling β_{120} enters the regressions as a control
A9	Truncated CARs (delistings valued at last price)	HOSTAGE permanence	Bounded, direction stated — optimistic for HOSTAGE; trunc% printed per archetype, so permanence is <i>understated</i> , strengthening that finding
A10	Two-way (stock × month) clustering captures the error dependence	all panel t-stats	Maintained — conservative relative to day-clustering; overlap additionally attacked head-on by the non-overlap subsample (§13B)
A11	Estimated archetype labels → attenuation, not inflation	dummy coefficients	Maintained — classical errors-in-dummies logic; effects understated
A12	INNOV's full-sample fit is a conservative competitor	§14 T3	Argued + bounded — hindsight strengthens the <i>alternative</i> , so survival is the stronger claim; INNOV's own T2 finding carries the decomposition-only caveat
A13	FII-slice PIN and FII-flow λ proxy their all-market versions	§16–17	Maintained, stated limitation — the endorsement is relative loadings within one consistent slice

#	Assumption	Where it bites	Status
A14	15 bps one-way is the relevant cost benchmark	§18 verdicts	Bounded — the full 0/5/10/15/30 grid is reported (outputs/metrics/backtest_metrics, conclusions stated as breakevens, not point verdicts)
A15	The liquid-946 universe represents FII price impact	external validity	Measured (Module 14) — the reversal does <i>not</i> generalize to the attributable tail (no-effect verdict, §13.6); tail dislocations are larger and unrecovered. The reversal claim is a liquid-market claim by measurement, not by scope restriction

20. Why this research is valuable, audience by audience

The running “value” thread, consolidated — with the honest calibration first: **the theory here is not new**. Temporary-versus-permanent impact of liquidity versus informed trading dates to Kraus–Stoll (1972) and runs through the entire block-trade and fire-sale literature. Pitched as a conceptual discovery, this work would be rightly rejected. Its contributions are measurement, evidence, and method — and after the Module-13 referee tests, they rank in this order:

1. **The identifier audit** (§2.2) — a standalone methodology for any masked-panel dataset, and the part an external reviewer independently singled out as most original.
2. **The participant-composition measure**, with its incremental information content now quantified: +28% relative IC ($t = 3.09$) over the standard feature set, robust to direct flow-magnitude controls, tail-concentrated by three independent tests (§13.5).
3. **The transitory/permanent evidence** — model-class-independent (§5.6), passive-mechanics-excluded, PIN-corroborated.
4. **The HMM**, demoted by our own ablation to what it is: a taxonomy convenience that buys episode smoothness.

By audience:

For empirical microstructure researchers. The first (to our knowledge) daily-frequency, entity-composition-based identification of the transitory/permanent split in an emerging market — built from a data class (masked depository records) that normally never leaves regulators. Plus two reusable negative results: single-day concentration snapshots are structurally invisible to HMMs (§5.4, with the autocorrelation diagnosis and audit-legal smoothing repair), and disclosed block-deal data measures visibility thresholds, not institutional concentration (§11.1).

For anyone who must trade Indian equities. The backtests locate the value precisely: not a standalone

strategy (breakevens 2–8 bps), but free information for agents whose costs are already sunk. Concretely: a desk liquidating a position into a volume-marked concentrated-selling episode is selling into the one configuration that reverts +49–65 bp within a month; a PM watching a holding decline *quietly* on dispersed FII selling is watching the configuration that never comes back, and that an independent PIN estimator flags as $\sim 3\times$ more informed.

In money: a desk instructed to sell Rs.100 crore of a mid-cap that is currently inside a concentrated-selling episode gives up roughly the episode-end reversal — $\sim 37\text{--}49$ bp out-of-sample (bounce-free to standard window, §13.5b) — by executing immediately rather than after the concentrated flow stops: **$\approx$ Rs.37–49 lakh on that one order**, at zero incremental cost, from a screening variable computable each evening from depository data. Statistical readers remember $t = 2.95$; desks remember fifty lakh. The permanent side prices identically in reverse: treating a quiet, dispersed-selling decline as “noise that will come back” is, on this evidence, a decision to hold the one configuration that does not.

For risk and policy. The composition lens inverts the usual panic ordering: headline-grabbing concentrated FII dumps are the self-correcting kind; quiet, broad, low-volume FII exits are the informative kind. A regulator monitoring FII outflows gets a screening variable that separates the two in real time from depository data it already possesses.

For methodologists and students. The project is a worked example of pre-registered, gate-driven empirical research on messy financial data: every claim has a falsification test that could have fired, five of them did fire (§21), and the two headline “discoveries” are both *survivals of attack* rather than products of search. The repository reproduces everything from raw data in ~ 8 minutes (python `pipeline.py --all`).

For the ML-in-finance conversation. A documented case where the fashionable escalation (GBT $\rightarrow$ LSTM) was evaluated under pre-registered bars and correctly *declined*: the flexible learner’s edge decomposed into regime information plus static characteristics plus a sub-significance timing residual (§15.4). Knowing when the model class is not the bottleneck is worth more than another model.

21. The failure log: everything that broke and how we knew

The project’s quality argument is not that nothing broke — eleven things broke — but that every break was caught by a gate, diagnosed read-only, fixed at the root, and recorded. Full narrative: `docs/research_log/FII_Module5_validation_log.md`; summary:

#	Failure	Class	Caught by	Fix
1	Assumed NSE prev_close is CA-restated (it is raw)	false assumption	Gate 0: ex-day medians identical raw vs prev-close	built the CA factor pipeline (§8.4)
2	Combined split+bonus rows parsed as bonus-only (TITAN, ONGC...)	parser bug	tape-verification disagreements clustering at $\times 5/\times 10$	keyword-anchored split regex + dedupe guard

#	Failure	Class	Caught by	Fix
3	Pasted Colab cell silently lost “-1” in the return formula (all 5.94M rows)	execution $\neq$ file	internally contradictory printout $\rightarrow$ read-only diagnostic found ret_cc = fresh+1.0 exactly	byte-exact script execution, ancestor of this repo’s runner
4	Symbol migrations put CA events on chains without the crash (+2860% fabrications)	data-class trap	application guard: post-adjustment ret >50%	guard nulls & logs (45 catches, all renames)
5	v1 fit-cap used most-recent-400 rows (recency-biased emissions)	subtle sampling bias	design review in 3A	random contiguous blocks
6	GMM threshold calibration produced a 49/51 coin-flip split (boundary +0.155)	model failure	its own pre-registered stability test ($\Delta=0.665$) + k-means & test-GMM disagreement	quantile rule; three-way convergence at -0.513
7	Event-study bootstrap tested vs zero under a +52bp drifting baseline (placebo “significant”)	inference flaw	ROBOT row significance	difference-in-differences vs ALL_LABELED baseline
8	TRAIN regression n collapsed 66k $\rightarrow$ 27k (index-return nulls poisoning rolling betas)	silent attrition	n printed per spec	zero-fill 0.6% index gaps; coefficients unchanged — never selection
9	Single-feature IC table lacked per-day dropna (F_breadth “-0.135” artifact)	diagnostic bug	implausibility check on one number	recomputed (+0.012); GBT itself unaffected
10	ROBOT placebo failed in TEST	control failure	flagged, then decomposed	structural (END = directional transition); HOSTAGE promoted as the valid null

#	Failure	Class	Caught by	Fix
11	Re-run calibration silently regenerated the rejected GMM thresholds (the falsification was manual in Colab)	provenance gap	artifact fingerprint diff on the census	falsification rule made binding in code (§6.4)
12	Tail-label build: probit NaNs passed <code>is_not_null</code> (polars NaN $\neq$ null), poisoning the quantile cuts — backbone printed an impossible +nan cut, BUY regime silently empty, census deformed into a false gate breach	library semantics trap	internally contradictory printout (a probit $q(0.29)$ cannot be +0.29)	<code>fill_nan(None)</code> at the transform + assert that cuts are finite and straddle zero (§13.6)

Two wrong economic narratives also died in the open: informed-Shark / forced-Hostage (§10.3) and block-visibility (§11.1). The narrative that survived did so against a pre-committed stopping rule.

22. Conclusion

Fourteen years of masked FII transaction data, one audited measurement (within-day participation concentration), one deliberately blind regime model, and a validation gauntlet that killed its authors' hypothesis twice, yield two regularities:

1. **Concentration separates transitory from permanent FII price impact.** Concentrated episodes are liquidity events — volume-marked at the climax, reverting +49–65 bp over the following month (+37–58 bp with the first two days excluded, §13.5b), robust to fixed effects, two-way clustering, loser-reversal controls, overlap, horizon, dose–response, and a hindsight-advantaged flow-surprise control. Dispersed selling is the opposite in every measured respect — quiet, permanent, and carrying $\sim 3\times$ the informed-trading probability by an estimator that never saw a price.
2. **The surprise component of FII flow reverts** out of sample — a decomposition-level regularity, stated with its look-ahead caveat permanently attached.

The regularities are economically thin by design of the world: breakeven trading costs of 2–8 bps explain why they persist un-arbitraged, and locate their use with agents whose trading costs are already paid. The model itself was shown *not* to be the binding constraint — a flexible learner with more features could not clear a pre-registered dynamics bar — so the honest frontier is more data (the illiquid tail, where the permanent-impact mechanism should be strongest), not more model.

What we would defend most, if allowed one claim: not the 65 basis points, but the *procedure* — pre-registered predictions, one gated test per step, read-only diagnosis before any fix, falsification rules that bind in code, and a repository where every number in this document regenerates from raw data in eight minutes. The finding inverted its own hypothesis and survived. That is what the procedure was for.

— *End.*